

Modelling Financial Markets

Chapter 2: Market Efficiency — From EMH to Adaptive Markets



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One question, five steps

- Can information available today improve a decision after risk and costs?
- Core: follow the five steps and answer their checkpoints.
- Practice: justify your interpretation using the examples and seminar.
- Further study: select a derivation, another test or a research extension.

- 1. Returns and information
- 2. From a pattern to a test
- 3. A forecast must work on new data
- 4. How prices react to news
- 5. Which conclusions survive scrutiny?
- Topics

1. Returns and information

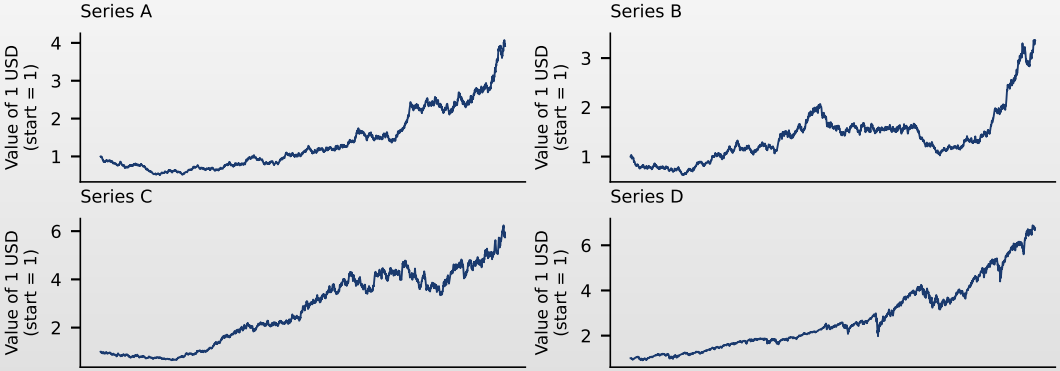
- **RW1**: Random Walk 1 — independent, identically distributed increments.
- **RW2**: Random Walk 2 — independent increments, possibly with different distributions.
- **RW3**: Random Walk 3 — uncorrelated increments, possibly dependent.
- Compute a return and state the information available before the trade.
- Distinguish RW1, RW2 and RW3 using their assumptions on returns.
- Explain why rejection also depends on the risk model.
- Contents

- A price rises from 100 to 105 without a dividend: the simple return is $105/100 - 1 = 5\%$.
- The log return is $\ln(105/100) \approx 0.04879$, or 4.879%.
- A total return also includes dividends: with a dividend of 2, $(105 + 2)/100 - 1 = 7\%$.
- A random walk adds an unpredictable innovation to the previous value
  - independent coin-toss increments of $+1$ or -1 give a simple example.

A Game: Which Series Is the Real Market? (1/2)

S&P 500: Standard & Poor's 500.

One of these is the S&P 500 (2010-2026); three are simulated random walks with the same drift and volatility



MFM_ch2_random_walk_game

A Game: Which Series Is the Real Market? (2/2)

- One panel is the S&P 500 (2010–2026); three are random walks with the same mean and volatility. Vote: A, B, C or D?

MFM_ch2_random_walk_game

A Game: Which Series Is the Real Market? (2/2)

- One panel is the S&P 500 (2010–2026); three are random walks with the same mean and volatility. Vote: A, B, C or D?
- Answer: **D**. Roberts (1959): random walks produce "patterns" by chance; we need statistics, not eyes
- Bachelier (1900) and Samuelson (1965): properly anticipated prices fluctuate randomly, so unpredictability is the mark of a well-functioning market

MFM_ch2_random_walk_game

Risk adjustment and information: examples

- An equity premium is the expected stock return above the risk-free return: $8\% - 3\% = 5$ percentage points in an illustrative year.
- Risk adjustment compares a strategy with a benchmark bearing comparable risk, after trading costs.
- A 10% expected return against an 8% risk benchmark leaves 2 percentage points before costs
  - a realised 10% in one year does not establish that expectation.
- Past prices belong to the weak-form information set
  - a published earnings report adds public information
  - unpublished earnings are private information.

Efficiency and the Information Set

- Fama (1970): a market is **efficient** if prices "fully reflect" all available information $\mathcal{I}_t$; portrait and surveys: Appendix
  - formally: no strategy using $\mathcal{I}_t$ earns **expected** returns above the fair (risk-adjusted) return
  - precursors Bachelier (1900) and Samuelson (1965), with their two results: Appendix

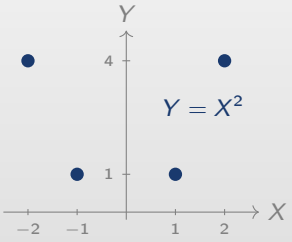
Form	Information set $\mathcal{I}_t$	Typical test (this chapter)
Weak	Past prices and returns	Autocorrelation, runs, variance ratios, long memory
Semi-strong	All public information (news, reports)	Event studies
Strong	Public and private (insider) information	Returns of insiders and fund managers

- Nested: strong $\Rightarrow$ semi-strong $\Rightarrow$ weak; risk-adjusted returns with factor models follow in Chapter 3

Correlation and dependence in returns

- Lag-k autocorrelation is the correlation between a return and the return k periods earlier.
- Correlation measures linear association: positive for movements together, negative for opposite movements.
- Zero correlation excludes linear association, but may leave dependence in squared returns.

- Toy example: $X = -2, -1, 1, 2$, equally likely; $Y = X^2$.
- The mean of X is zero; the mean of XY is $(-8 - 1 + 1 + 8)/4 = 0$. Covariance, hence correlation, is zero.
- Yet knowing X determines Y exactly: they are dependent.



Conditional means and martingales: an example

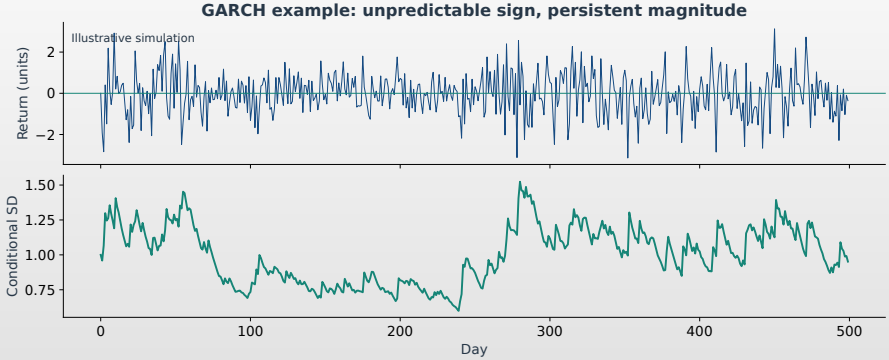
- A conditional mean averages possible next outcomes using the probabilities implied by what is known today.
- From a current value of 100, equally likely values 110 and 90 have conditional mean $0.5(110) + 0.5(90) = 100$.
- If this equality holds at every date given all past information, the process is a martingale
  - its next increment has conditional mean zero.
- With a required return of 5%, equally likely returns 15% and -5% form a fair game after subtracting 5%.

Independence, volatility clustering and GARCH

- **GARCH**: Generalized AutoRegressive Conditional Heteroskedasticity.
- **i.i.d.**: independent and identically distributed.
- Independent observations give no information about one another
  - identically distributed observations share the same distribution.
  - Together: i.i.d.
- Volatility clustering means that large absolute moves cluster in time, regardless of their sign
  - for example $+3, -4, +5\%$ after quiet days near $\pm 0.2\%$.
- Heteroskedasticity is changing variance
  - homoskedasticity is constant variance under the stated conditioning information.
- GARCH models tomorrow's conditional variance from past squared shocks and variances
  - returns can be uncorrelated while their magnitudes remain dependent.

Volatility clustering in returns

SD: Standard Deviation.



Illustrative GARCH: large absolute returns occur in groups. The lower panel shows the conditional SD, not a forecast of the return sign.

RW1, RW2, RW3: the restrictions on returns

$$r_t = \ln P_t - \ln P_{t-1} = \mu + \varepsilon_t$$

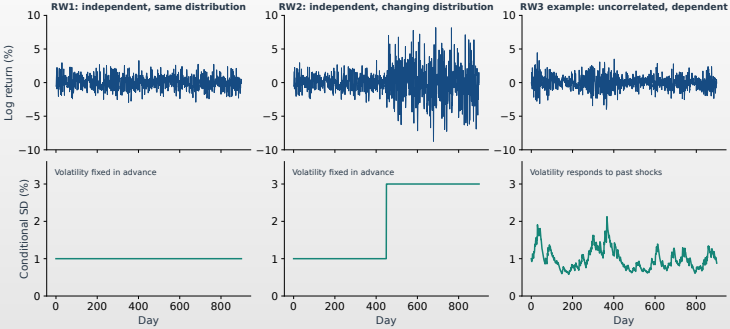
- μ: a constant mean; ε_t: the zero-mean innovation. The random walk concerns log prices; the restrictions concern their increments, the log returns.

	Requirement	Illustrative return model
RW1	Independent, identical distribution	$r_t = \mu + 0.01 z_t$, z_t i.i.d. $N(0,1)$
RW2	Independent, distribution may vary	$r_t = \mu + \sigma_t z_t$, σ_t = 1% then 3%, fixed in advance
RW3	Uncorrelated, possibly dependent	GARCH: conditional variance responds to past squared shocks

RW1 $\Rightarrow$ RW2 $\Rightarrow$ RW3

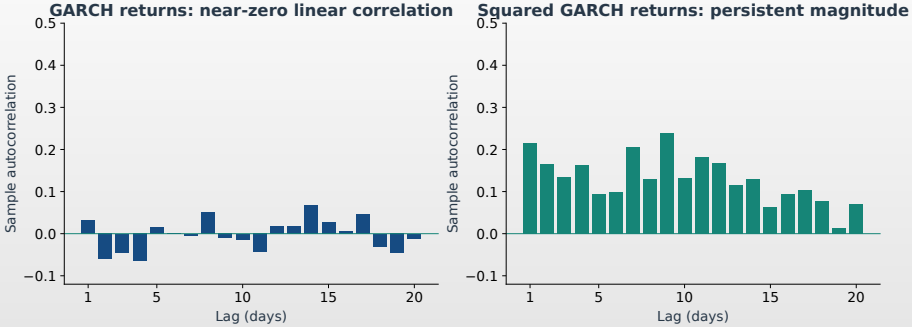
- RW1 to RW2: remove the identical-distribution restriction, retain independence.
- RW2 to RW3: retain zero linear correlation, allow dependence. The implications require the relevant moments to exist.
- The examples distinguish the classes: a predetermined variance shift violates RW1; GARCH dependence violates RW2.
- RW3 does not imply an unpredictable conditional mean: nonlinear predictability can remain. GARCH with zero-mean independent innovations is one example, not the definition of RW3.

Three models for returns and volatility



- Illustrative models with $\mu = 0$.
- RW2 volatility is fixed in advance
- GARCH volatility responds to observed shocks.
- Visual clusters alone do not prove dependence.

GARCH: returns versus squared returns



- Sample correlations fluctuate around zero for returns
- positive squared-return correlations reveal dependence in magnitude.
- These are illustrative sample estimates, not hypothesis tests.

CAPM and a joint hypothesis: example

CAPM: Capital Asset Pricing Model.

- CAPM uses market beta, the sensitivity to market returns, to set a required return
  - Chapter 3 develops the model.
- If a strategy has expected return 10% and CAPM requires 8%, the apparent abnormal return is 2 percentage points.
- If omitted risk actually requires 10%, the apparent anomaly disappears: rejection can concern the risk model as well as market efficiency.

The Joint-Hypothesis Problem

- "Fair return" requires a model of expected returns (e.g. CAPM, Chapter 3)
- Every test of efficiency is a **joint** test of efficiency and of the asset-pricing model Fama (1991)
  - A rejection may mean an inefficient market **or** a wrong model of risk
  - Predictable **expected** returns are allowed if they compensate time-varying risk
- Some investors beat the market by luck: the question is whether success persists
- At daily horizons expected returns are tiny, so the choice of model matters little: weak-form tests are the cleanest
- Efficiency can never be "proved", only not rejected by a given test on given data

Check: returns and information

- A price rises from 100 to 105 without dividends. Which return is 5%?
- Can independent returns have SD fixed at 1% then 3% in advance?
- Does zero autocorrelation establish independence?
- Check the reasoning

Answers: returns and information

- 5% is the simple return; the log return is about 4.879%.
- Yes: a predetermined SD change is compatible with RW2, but violates RW1.
- No: the GARCH example has dependent magnitudes despite zero return autocorrelation.
- Continue to the next step

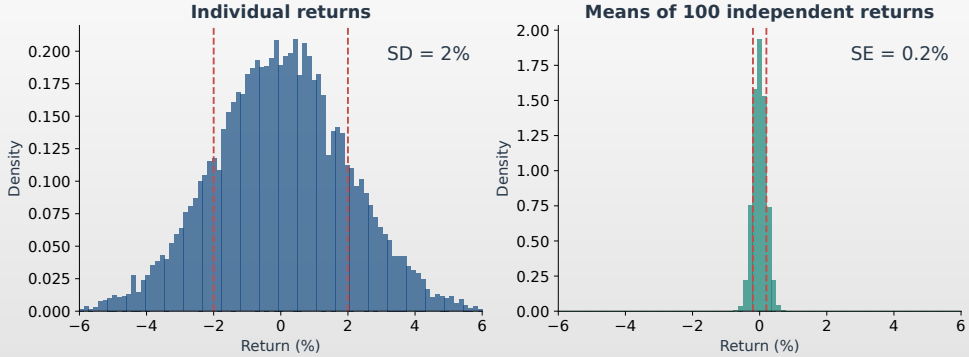
2. From a pattern to a test

- Read the sign, uncertainty and economic size of an autocorrelation.
- Explain why volatility clustering changes inference.
- Read a variance-ratio profile without inferring a trading profit.
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Dispersion and uncertainty

- Standard deviation (SD) measures how dispersed the returns are.
- Standard error (SE) measures how much an estimate varies across repeated samples.
- For 100 i.i.d. returns with SD 2%, the sample mean has SE $2\%/\sqrt{100} = 0.2\%$.
- A null band shows the estimates typically generated under a specified null hypothesis.

SD and SE on the same horizontal scale



Illustrative Normal samples: individual returns have SD 2%; means of 100 independent returns have SE 0.2%.

Reading a statistical test

- H_0 is the null being tested, such as zero autocorrelation
  - H_1 is the alternative, such as nonzero autocorrelation.
- A p-value is the probability, under H_0 and the test assumptions, of a statistic at least as extreme as observed
  - it is not the probability that H_0 is true.
- At significance level $\alpha = 5\%$, $p = 0.03$ leads to rejection
  - $p = 0.20$ does not.
  - A valid level controls false rejection when H_0 holds.
- Power is the rejection probability at a specified alternative: detecting an effect in 80 of 100 simulated samples means estimated power 80%.

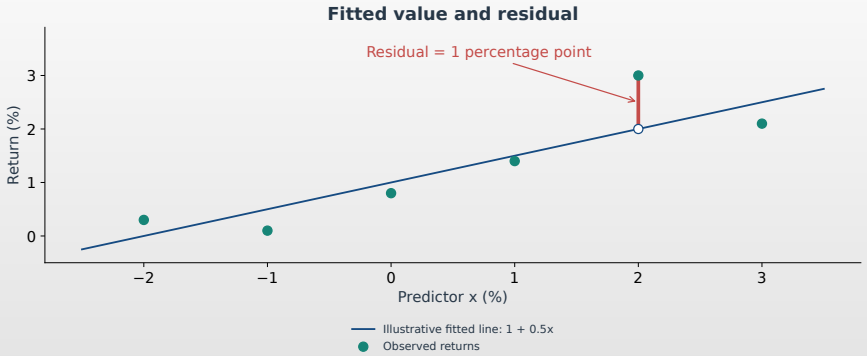
Autocorrelation: what to read first

- Autocorrelation compares a return with a previous return at a stated lag.
- Positive: persistence; negative: reversal. Neither sign establishes profit.
- The estimate describes the effect size; its standard error describes uncertainty.
- Use robust inference when squared returns are dependent.
- Estimator and robust standard error

OLS, residuals and persistence

- **OLS**: Ordinary Least Squares.
- **AR**: AutoRegressive model.
- OLS chooses regression coefficients to minimise the sum of squared residuals
  - a residual is observed minus fitted return.
- Example: fitted return $1\% + 0.5x$ equals 2% at $x = 2\%$
  - an observed 3% leaves a residual of 1 percentage point.
- AR(1) relates a variable to its previous value
  - with coefficient 0.9 a unit shock still contributes $0.9^2 = 0.81$ two periods later.
- A unit root means coefficient 1 in this AR(1): shocks do not decay.
  - A coefficient close to 1 produces persistent predictors and difficult inference.

Reading a regression line



At x = 2%, the illustrated line predicts 2%; the observed return is 3%, leaving a residual of 1 percentage point.

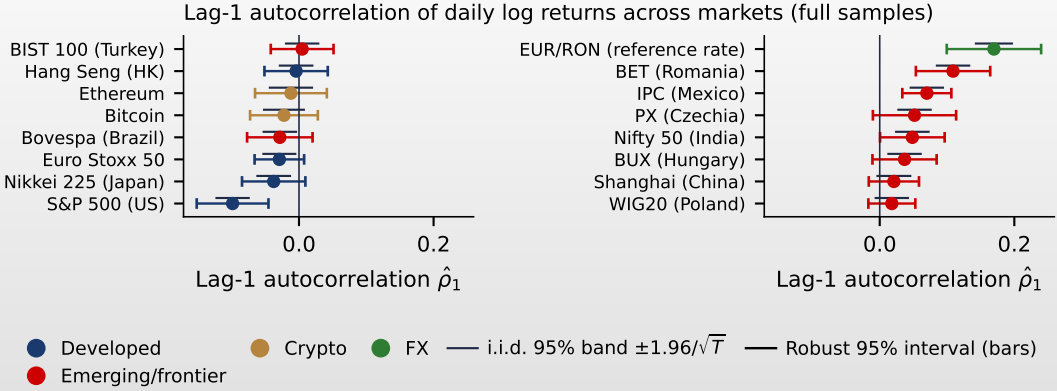
Interpreting dependence and R^2

- R^2 is the share of sample variation explained by a fitted regression
  - in a simple regression with an intercept it equals the squared sample correlation.
- A correlation of 0.10 corresponds to $R^2 = 0.01$, or 1% explained variation: statistical detection can coexist with weak prediction.
- Thin trading leaves stale prices: news today may enter an infrequently traded stock's recorded return only tomorrow.
- Liquidity provision means accepting others' trades
  - temporary price concessions can reverse.
  - Managed float allows exchange-rate adjustment with central-bank intervention.

Markets and data sources

- **US**: United States.
- **HK**: Hong Kong.
- **BIST**: Borsa İstanbul (BIST 100 index) — the Istanbul Stock Exchange (BIST 100 index).
- **WIG20**: Warszawski Indeks Giełdowy 20 — the Warsaw Stock Exchange index of the 20 largest companies.
- **BUX**: Budapesti Értéktőzsde index — the Budapest Stock Exchange index.
- **PX**: Prague Stock Exchange index (index Burzy cenných papírů Praha).
- **IPC**: Índice de Precios y Cotizaciones — the Mexican Stock Exchange index.
- **BET**: Bucharest Exchange Trading (index).
- **FX**: Foreign eXchange.

Lag-1 Autocorrelation Across Markets (1/2)



MFM_ch2_autocorrelation

Lag-1 Autocorrelation Across Markets (2/2)

- Dots: $\hat{\rho}_1$ with robust 95% intervals; thin segments: the naive i.i.d. band; colours: the market group (FX, foreign exchange: EUR/RON)
- Robust intervals use $\sqrt{\hat{\tau}_1}$ instead of $1/\sqrt{T}$ and are wider when large moves cluster: S&P 500 SE 0.027 (robust) vs 0.012 (i.i.d.)
- Positive and significant: BET 0.109 (SE 0.028), IPC 0.070, EUR/RON 0.170
- Negative and significant: S&P 500 -0.099
- Bitcoin -0.022, Ethereum -0.012: not significant
- Candidate explanations: thin trading (BET), a managed exchange rate (EUR/RON), short-term overreaction and liquidity provision (S&P 500)
- Economic size: $R^2 = \hat{\rho}_1^2$, i.e. 0.97% (S&P 500) and 1.19% (BET) of the daily variance

MFM_ch2_autocorrelation

Stationarity and overlapping returns

- Covariance stationarity means a constant mean, finite constant variance, and covariance depending only on the lag, not the calendar date.
- A stable AR(1), $x_t = 0.5x_{t-1} + u_t$, started in its stationary distribution with i.i.d. finite-variance shocks, is an example
  - a permanent mean shift violates the definition.
- Overlapping five-day returns use days 1–5, then 2–6: four days are shared, so the sums are dependent even with independent daily returns.
- Leverage effect in volatility means a negative shock raises future volatility more than an equally large positive shock
  - it differs from borrowing to buy an asset.

The Variance Ratio: Idea

VR: Variance Ratio.

- Uncorrelated increments with constant variance: variance grows linearly with the horizon, $\mathrm{Var}(r_t + \cdots + r_{t-q+1}) = q\,\mathrm{Var}(r_t)$
  - q: the horizon in days (e.g. $q = 5$ is one trading week)
  - With changing variances (RW2, RW3) the variance of the sum is the sum of the variances; the sample $\widehat{\mathrm{VR}}(q)$ still tends to 1 under the conditions of Lo & MacKinlay (1988)
- **Variance ratio**: $\mathrm{VR}(q) = \dfrac{\mathrm{Var}(r_t + \cdots + r_{t-q+1})}{q\,\mathrm{Var}(r_t)}$, equal to 1 for any covariance-stationary uncorrelated series
- For a covariance-stationary series, a weighted sum of autocorrelations: $\mathrm{VR}(q) = 1 + 2\sum_{k=1}^{q-1}\left(1 - \frac{k}{q}\right)\rho_k$
- $\mathrm{VR}(q) > 1$: momentum (positive autocorrelation); $\mathrm{VR}(q) < 1$: mean reversion
- Worked check: S&P 500 $1 + \hat{\rho}_1 = 0.901$ vs estimated $\mathrm{VR}(2) = 0.901$

Which variance-ratio horizons do we test?

- Lo & MacKinlay (1988), Table 1a: four horizons, $q = 2, 4, 8, 16$, with weekly observations.
- Here the four test horizons are $q = 2, 5, 10, 20$ days: two days, about one week, two weeks and one trading month for equities.
- The curve through $q = 60$ is descriptive. Its band is pointwise: uncertainty is assessed separately at each horizon.
- Four horizons are a stated design choice, not a guarantee of sufficiency.
- The Chow–Denning test combines the chosen horizons into one joint rejection decision Chow & Denning (1993).

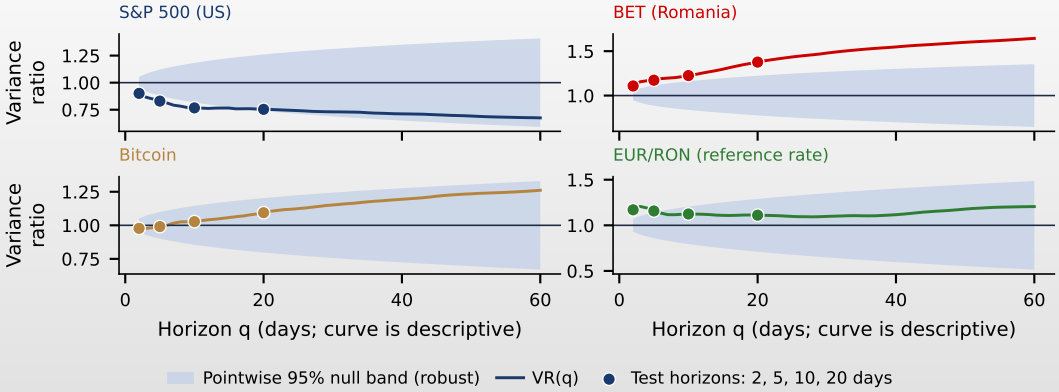
Reading a robust variance-ratio band

- The benchmark $\mathrm{VR}(q) = 1$ holds for uncorrelated increments even when their conditional variance changes, under the stated RW3 conditions; the robust band is built for this null.
- $\hat{\theta}(q)$ estimates the variance of $\sqrt{T}\,(\widehat{\mathrm{VR}}(q) - 1)$; when large moves follow large moves it grows, and the band widens.
- The band is pointwise: it concerns one horizon at a time, not all horizons jointly.
- Formula and validity conditions

Why the robust band is wider: a numerical check

- For $q = 2$: $z = \sqrt{T}\,(\widehat{\mathrm{VR}}(2) - 1)$ under RW1, and $z^* = \sqrt{T}\,(\widehat{\mathrm{VR}}(2) - 1)/\sqrt{\hat{\theta}(2)}$, with $\hat{\theta}(2) = \hat{\delta}_1$.
- $\hat{\delta}_1$ converges to 1 under the i.i.d. null and grows when large moves follow large moves.
- Toy numbers: $T = 400$, $\widehat{\mathrm{VR}}(2) = 0.90$, so $z = 20 \times (-0.10) = -2.0$: a rejection at 5%.
- If clustering gives $\hat{\delta}_1 = 4$: $z^* = -2.0/\sqrt{4} = -1.0$, no rejection; the band around 1 is twice as wide.
- S&P 500, same formulas: $z = -8.12$ but $z^* = -3.64$: smaller in absolute value, yet still a rejection.

Variance-Ratio Profiles (1/2)



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Variance-Ratio Profiles (2/2)

- $\widehat{\mathrm{VR}}(q)$ for $q = 2, \ldots, 60$; shaded: robust 95% band around 1, from $\hat{\theta}(q)$, built for RW3 (uncorrelated increments whose variance may change)
- S&P 500 at the four tested horizons: reversal, outside the band at $q = 2, 5, 10$, inside at $q = 20$
  - $\widehat{\mathrm{VR}}(2) = 0.901$, $\widehat{\mathrm{VR}}(5) = 0.830$, $\widehat{\mathrm{VR}}(10) = 0.767$, $\widehat{\mathrm{VR}}(20) = 0.753$
- BET at the same horizons: persistence accumulating with q, outside the band at all four
  - $\widehat{\mathrm{VR}}(2) = 1.108$, $\widehat{\mathrm{VR}}(5) = 1.173$, $\widehat{\mathrm{VR}}(10) = 1.225$, $\widehat{\mathrm{VR}}(20) = 1.377$
- $q = 60$, about three trading months, is a descriptive extension: it is not one of the four tested horizons
  - S&P 500: $\widehat{\mathrm{VR}}(60) = 0.674$; BET: $\widehat{\mathrm{VR}}(60) = 1.644$
- Bitcoin: inside the band up to about one month; EUR/RON: short-run persistence of a managed float
- The band widens with q: a curve can point in one direction without being significant; direction is not evidence

MFM_ch2_variance_ratio

Check: patterns and uncertainty

- ⊡ Why can an i.i.d. band be misleading under volatility clustering?
- ⊡ Does positive autocorrelation imply profit after costs?
- ⊡ What does a variance ratio below one suggest?
- ⊡ Transfer: the VR estimate stays unchanged but its standard error doubles. What happens to a statistic of magnitude 2.4 and a two-sided 5% decision?
- ⊡ Check the reasoning

Answers: patterns and uncertainty

- The i.i.d. standard error ignores dependence in squared returns.
- No: effect size, risk, implementation and costs still matter.
- It suggests reversal over the chosen horizon; significance needs valid inference.
- The statistic halves to 1.2; it no longer crosses 1.96. The estimated effect is unchanged; its evidence is weaker.
- Continue to the next step

3. A forecast must work on new data

- Keep estimation data separate from forecast evaluation.
- Compare squared errors with those of a stated benchmark.
- Separate predictive accuracy from profit after risk and costs.
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What information enters a forecast?

- A predictor is an observable used to forecast a future return.
- Here dp is the log dividend–price ratio; dividends are cash payments to shareholders.
- The dividend data are interpolated: this example is pseudo-out-of-sample, not fully real-time.
- Regression estimation and bias

The essential out-of-sample comparison

OS: Out-of-Sample.

- $R^2_{OS} = 1 - \sum_t (r_t - \hat{r}_t)^2 / \sum_t (r_t - \bar{r}_t)^2$.
- $\hat{r}_t$: model forecast; $\bar{r}_t$: historical-mean forecast using the same available data.
- Positive score: smaller squared errors than the benchmark on the evaluation sample.
- A positive score alone does not establish statistical significance or profit.
- Clark–West formula and estimation details

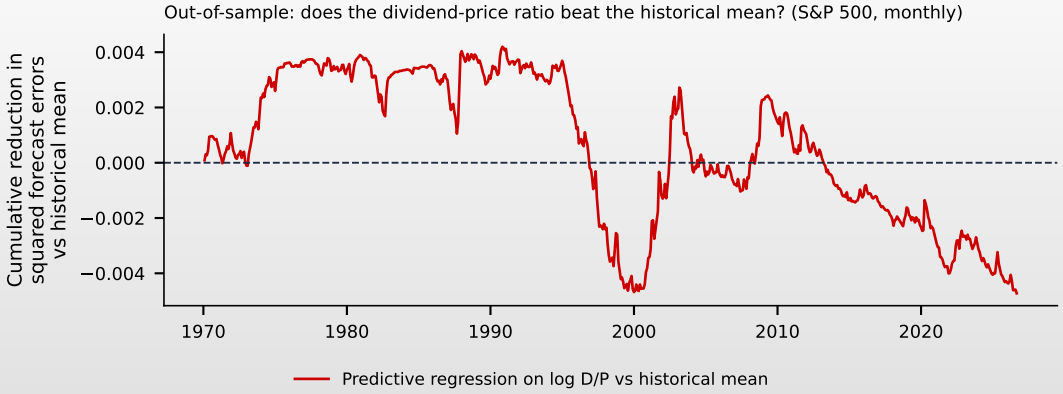
Forecast evaluation: a worked example

- In-sample evaluates the observations used for estimation
  - out-of-sample evaluates observations excluded from estimation.
- MSPE is the mean squared prediction error: errors of 1 and 3 percentage points give $(1^2 + 3^2)/2 = 5$ squared percentage points.
- If benchmark MSPE is 4 and predictor MSPE is 3, $R^2_{OS} = 1 - 3/4 = 25\%$: squared errors fall by one quarter.
- Nested models contain one another: setting the slope to zero reduces an intercept-plus-predictor model to an intercept-only model.

How the forecasts are formed

- An expanding window re-estimates the model each month using only observations before the month forecast.
- The benchmark uses the historical mean of the same available observations.
- Predictive regressions and bias adds the Clark–West test and Campbell–Thompson restrictions.

The Dividend–Price Ratio: Out-of-Sample Forecasts



MFM_ch2_predictability

Reading the cumulative squared-error chart

- Each month t from 1970 adds $(r_t - \bar{r}_t)^2 - (r_t - \hat{r}_t)^2$: the squared error of the historical mean minus that of the *dp* forecast; the curve is the running sum, in squared fractional log-return units.
- A rising segment: in those months *dp* forecast better than the historical mean; a falling segment: worse. A fall is a loss of forecast accuracy, not a trading loss.
- The last value is $\sum_t (r_t - \bar{r}_t)^2 - \sum_t (r_t - \hat{r}_t)^2$, so it has the sign of R^2_{OS}: above zero, $R^2_{OS} > 0$; below zero, $R^2_{OS} < 0$.
- If a curve ended at -0.002, the *dp* forecasts would have accumulated 0.002 more squared error than the benchmark: $R^2_{OS} < 0$.
- Forecast errors, in percentage points: historical mean $2, 1, 3$; *dp* forecast $1, 2, 3$.
- Monthly terms $4 - 1 = 3$, $1 - 4 = -3$, $9 - 9 = 0$: the curve rises to 3, falls back to 0 and stays there.
- Both sums of squared errors are 14, so $R^2_{OS} = 1 - 14/14 = 0$: neither forecast wins on this sample.
- Results for the S&P 500 curve: R^2_{OS}, the Clark–West test and the 1990s episode

Check: forecasting on new data

- Which data may enter a forecast made today?
- What does a negative out-of-sample score say about the benchmark?
- Why can excellent in-sample fit fail on new data?
- Transfer: can a strategy make money while its forecast has negative out-of-sample R-squared? Explain the two benchmarks.
- Check the reasoning

Answers: forecasting on new data

- Only information available by the forecast origin may enter estimation.
- The model has larger squared errors than the benchmark on that sample.
- The model may fit noise or exploit information unavailable at the forecast origin.
- Yes: negative out-of-sample R-squared means larger squared forecast errors than the historical mean. Positive profit can reflect market exposure; it does not establish superior forecasts or risk-adjusted profit after costs.
- Continue to the next step

4. How prices react to news

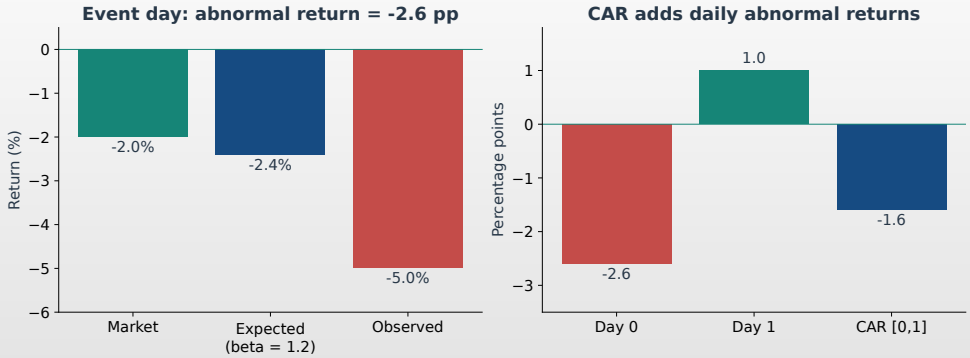
CAR: Cumulative Abnormal Return.

- Separate observed, expected and abnormal returns.
- Interpret CAR over a stated event window.
- Distinguish a measured reaction from proof of its cause.
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Event studies: the economic quantities

- A stock split changes the number of shares and price per share without mechanically changing wealth: one share at 100 becomes two at 50 in a 2-for-1 split.
- Market beta measures sensitivity: with intercept zero and beta 1.2, a market return of -2% predicts -2.4%.
- Abnormal return (AR) is the observed return minus the benchmark prediction: if the stock returns -5%, $AR = -5 - (-2.4) = -2.6$ percentage points.
  - In event studies AR means abnormal return, not the autoregressive model AR(p).
- CAR adds the AR values over an event window: -2.6 today and $+1$ tomorrow give -1.6 percentage points.

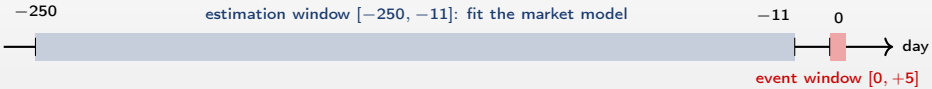
Abnormal return and CAR: an example



The benchmark explains part of the observed return. The remainder is the abnormal return (AR); CAR sums these remainders over the chosen days.

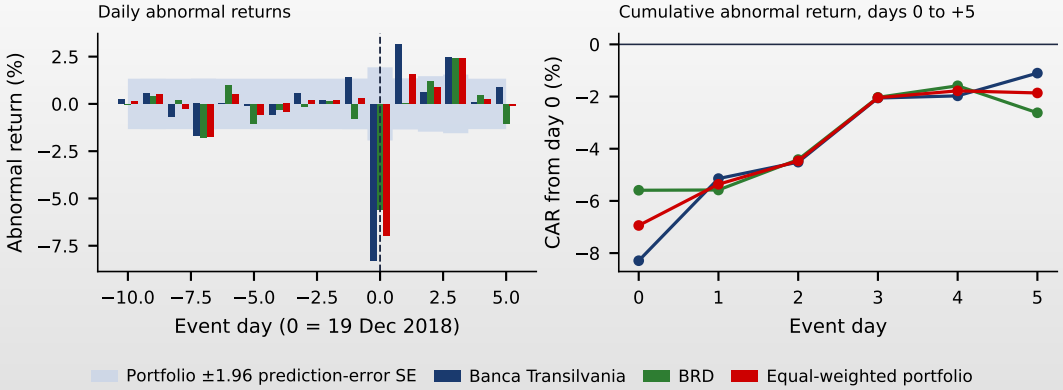
The Event-Study Method

AR (event study): Abnormal Return.



- Fama, Fisher, Jensen & Roll (1969): how fast do prices absorb public news (stock splits)?
- Market model: $r_{i,t} = a_i + b_i r_{m,t} + \varepsilon_{i,t}$, $r_{m,t}$ the market return; abnormal return $AR_{i,t} = r_{i,t} - \hat{a}_i - \hat{b}_i r_{m,t}$; cumulative abnormal return $CAR_i[\tau_1, \tau_2] = \sum_{t=\tau_1}^{\tau_2} AR_{i,t}$
- First-pass test: $t = CAR_i / (\hat{\sigma}_\varepsilon \sqrt{L})$, L the event-window length MacKinlay (1997)
  - a_i, b_i: intercept and market beta of firm i, fitted in the estimation window; $\hat{\sigma}_\varepsilon$: the residual standard deviation there; $[\tau_1, \tau_2]$: the days summed
  - AR: the return beyond what the market move predicts; $\sqrt{L}$: the SD of a sum of L independent residuals grows with $\sqrt{L}$
- For unanticipated news, efficiency predicts a jump on day 0 and no predictable drift or reversal afterwards
- Daily data can only show absorption within a day; speed measured in microseconds needs intraday data (Chapter 10); literature: ▸ Appendix

Worked Example: the Romanian Bank Tax of December 2018 (1/2)



MFM_ch2_seminar

Worked Example: the Romanian Bank Tax of December 2018 (2/2)

- Tax announced on the evening of 18 Dec 2018, day 0 = 19 Dec; Banca Transilvania (TLV), BRD and their equal-weighted portfolio vs BET (Seminar 2, B9; optional background: Seminar 0, B7)
- TLV: $\hat{b} = 1.18$, $\hat{\sigma}_\varepsilon = 0.83\%$, $AR_0 = -8.3\%$ ($t = -10.0$, first-pass formula)
  - Day 0 was also an extreme market day (BET -11.9%); allowing for it moves t to -6.85: event-study inference
  - $CAR[0,+5] = -1.1\%$ ($t = -0.54$, first pass), since $CAR[+1,+5] = +7.2\%$ ($t = 3.82$ with the prediction-error variance; the first-pass formula gives 3.87)
- Most of the day-0 fall reverses within the week: an **overreaction**, against semi-strong efficiency, or later news about the final form of the ordinance; one event cannot separate them

MFM_ch2_seminar

Check: reactions to news

- Observed return is -5%, expected return -2.4%. What is abnormal return?
- Why must you specify the event window?
- Could another announcement explain the same movement?
- Check the reasoning

Answers: reactions to news

- Abnormal return is -2.6 percentage points.
- CAR adds abnormal returns over the selected dates only.
- Yes: overlapping news is an alternative explanation, so the reaction alone is not causal proof.
- Continue to the next step

5. Which conclusions survive scrutiny?

- Account for the number of hypotheses searched.
- State what the evidence establishes and what it leaves open.
- Choose a check that could overturn your conclusion.
- Contents

Calendar effects and data snooping

- A calendar effect is a difference in mean returns associated with a day or month.
- Turn of the month (TOM): the last trading day of a month and the first three of the next month.
- A January effect compares January returns with the other months.
- Data snooping means selecting a pattern after searching the same data for many alternatives.

Reading the calendar table

HAC: Heteroskedasticity and Autocorrelation Consistent.

- The 15 tests combine five markets with three effects: weekday, January and TOM.
- HAC standard errors allow heteroskedasticity and autocorrelation; a t statistic divides an estimated effect by its standard error.
- One basis point (bp) is one hundredth of a percentage point.
- Regression specification and other anomalies

January and Turn-of-the-Month Effects

Market	January extra (%/month)	t_{HAC}	TOM extra (bp/day)	t_{HAC}	p
S&P 500 (US)	−0.66	−0.80	3.2	0.93	0.3539
BET (Romania)	3.48	1.49	17.3	3.62	0.0003
Euro Stoxx 50	−0.30	−0.29	5.1	1.21	0.2266
WIG20 (Poland)	0.53	0.40	8.2	1.73	0.0841
BUX (Hungary)	2.66	2.25	15.1	3.63	0.0003

- TOM: BET +17.3 bp per day and BUX +15.1 bp, both with $|t| > 3.5$; S&P 500 only +3.2 bp
- January: large but imprecise for BET ($t = 1.49$); BUX, with $t = 2.25$, is the only market with a January-effect p-value below 0.05
- We ran 15 tests (5 markets $\times$ 3 effects): which survive a correction for multiple testing?

Multiple testing: a small example

FWER: Family-Wise Error Rate.

- FWER is the probability of at least one false rejection in a family of tests
  - ten separate 5% tests do not generally control it at 5%.
- For three p-values $0.01, 0.03, 0.04$, Bonferroni compares each with $0.05/3 \approx 0.0167$: only the first rejects.
- Holm orders p-values: $0.01 < 0.05/3$ rejects
  - $0.03 > 0.05/2$ stops the procedure, so the remaining tests do not reject.
- The formula $1 - (1 - \alpha)^m$ for at least one false rejection assumes m independent true-null tests
  - dependence changes this probability.

Multiple Testing: Bonferroni and Holm

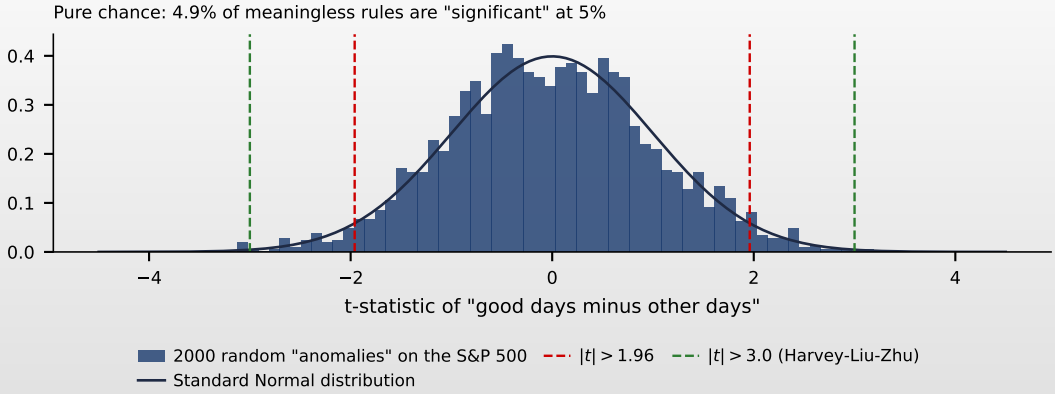
- **FWER** (family-wise error rate): the probability of at least one false rejection; with M independent tests at level α, FWER $= 1 - (1-\alpha)^M$
  - $M = 15$, $\alpha = 5\%$: FWER $= 53.7\%$, a coin flip
- **Bonferroni**: reject only if $p_i < \alpha/M$; guarantees FWER $\leq \alpha$ for any dependence
- **Holm**: sort $p_{(1)} \leq \cdots \leq p_{(M)}$; reject while $p_{(j)} < \alpha/(M-j+1)$, stop at the first failure
  - Same guarantee, never less powerful than Bonferroni
- Our 15 calendar tests: Bonferroni threshold 0.0033; 3 raw rejections, 2 survive Holm (BET and BUX turn of the month); Holm stops at BUX January
- Seminar B4 carries out the ranking step by step; bootstrap methods that use the dependence between tests: ▸ Appendix

Holm on the 15 calendar p-values

Market	Weekday, equal means	January	TOM
S&P 500 (US)	0.8263	0.4257	0.3539
BET (Romania)	0.3052	0.1368	0.0003
Euro Stoxx 50	0.6319	0.7738	0.2266
WIG20 (Poland)	0.2262	0.6865	0.0841
BUX (Hungary)	0.9327	0.0245	0.0003

- Two smallest: BET and BUX TOM, below $0.05/15 = 0.0033$ and $0.05/14 = 0.0036$: both rejected.
- Third smallest: BUX January, $p = 0.0245$, above $0.05/13 = 0.0038$: Holm stops. Below 0.05, it is the third raw rejection.
- Two-sided p-values; January and TOM from the HAC t with the Normal reference; weekday: Wald test of four equality restrictions (χ^2_4).

How Many "Anomalies" Does Pure Chance Produce? (1/2)



MFM_ch2_multiple_testing

How Many "Anomalies" Does Pure Chance Produce? (2/2)

- 2 000 meaningless rules on the S&P 500: each labels a random 20% of days as "good" and tests good minus other days
- 4.9% have $|t| > 1.96$, the 5% the test promises; the largest $|t|$ is 3.19
- Whoever tries many rules and reports the best is very likely to find a "significant anomaly"; data snooping in trading rules: Sullivan, Timmermann & White (1999)
- Harvey, Liu & Zhu (2016) suggest $|t| > 3.0$ for a new factor; here 0.25% of random rules pass. A higher bar is a heuristic, not a substitute for defining the family of tests tried

MFM_ch2_multiple_testing

Check: multiple tests and conclusions

- Why does testing many rules change the meaning of a small p-value?
- Does non-rejection prove efficiency?
- Which new sample or robustness check would you request?
- Check the reasoning

Answers: multiple tests and conclusions

- Searching many rules creates more opportunities for false rejection.
- No: limited power and the chosen alternative constrain the conclusion.
- A valid answer names a check and explains which weakness it addresses.
- Complete the core route

You have completed the core route

- State the information set, the null and the benchmark for every result.
- Explain uncertainty before making an economic claim.
- In the seminar, justify a calculation and challenge an interpretation.
- You can stop here on a first reading.
- Topics

Topics for further study

- ⊡ Data and conventions
- ⊡ Fair games, pricing and rational bubbles
- ⊡ Test construction and alternatives
- ⊡ Predictive regressions and bias
- ⊡ Long memory and changing efficiency
- ⊡ Event-study inference
- ⊡ Anomalies and behavioural explanations
- ⊡ Machine learning and research design
- ⊡ Synthesis and technical reference
- ⊡ Contents

Data Used in This Chapter

BNR: Banca Națională a României — the National Bank of Romania.

- 16 daily series up to 18 September 2026; sample sizes: Appendix
  - 13 equity indices from 2000: 4 developed (S&P 500, Euro Stoxx 50, Nikkei 225, Hang Seng), 9 emerging or frontier (BET, WIG20, BUX, PX, BIST 100, Bovespa, IPC, Nifty 50, Shanghai)
  - Bitcoin (from 2014; from 2011 in the yearly crypto analysis), Ethereum (from 2016) and EUR/RON, BNR reference rate (from July 2005)
- Daily log returns $r_t = \ln P_t - \ln P_{t-1}$, each series on its own calendar
  - P_t: the closing price (index level, exchange rate) on day t; ln: the natural logarithm
- Equity indices: weekdays only; a close equal to the previous close is removed, since it is usually a holiday filled with the previous price
  - the rule also removes the rare genuine zero-return day (S&P 500: 3 of about 6 700 days); the test statistics move by less than 0.05
- Crypto-assets trade 7 days a week: 365 observations per year
- Analyses that combine several series (e.g. event studies) align prices on common trading days before computing returns

Arbitrage, long and short positions

- How do fair games, valuation and rational bubbles fit together?
- Prerequisite: returns and information
- Topics
- Contents

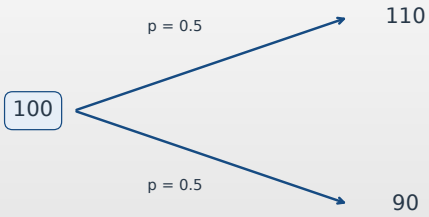
- Pure arbitrage is a strategy with no positive initial outlay with no possible loss and a positive gain in at least one possible state.
- Example ignoring costs: simultaneously buy an identical deliverable asset for 100 and sell it for 101
  - the locked-in difference is 1.
- A long position gains from a price rise
  - a short sale borrows and sells an asset, then buys it back: sale at 100, repurchase at 90 earns 10 before costs.
- Relative-value trading is risky when convergence is uncertain: a 20 price gap may widen to 30, triggering extra collateral (margin) and forced liquidation.

Discounting and the risk-neutral measure

- Discounting converts a future amount into today's units: 105 payable next year has present value $105/1.05 = 100$ at 5%.
- A risk-neutral measure uses pricing probabilities under which discounted traded prices, including dividends, are martingales in an arbitrage-free model.
- Example without dividends: stock 100, next-year prices 120 or 80, risk-free rate 5%
  - the pricing probability $q = 0.625$ gives $q\,120 + (1-q)80 = 105$.
- This q need not equal the actual probability of an increase: pricing probabilities are not forecasts of frequencies.

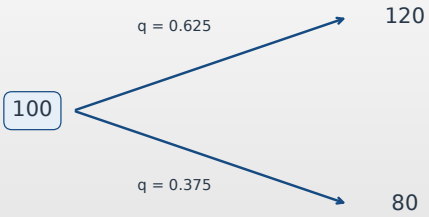
Probabilities and conditional means

Martingale under actual probabilities



Next-value mean = 100
Expected increment = 0

Risk-neutral pricing example



Next-value mean = 105
Discounted: 105 / 1.05 = 100

Left: the actual conditional mean stays at 100. Right: pricing probabilities give 105, whose discounted value is 100.

Fair games and martingales

- **Fair game** for excess returns: $\mathbb{E}[r_{t+1} - \mu_t \mid \mathcal{I}_t] = 0$
  - $\mathcal{I}_t$: the information available at the end of day t; $\mathbb{E}[\cdot \mid \mathcal{I}_t]$: the expectation given that information
  - μ_t: the required (equilibrium) return, the compensation for risk; in words: the return beyond μ_t cannot be forecast from $\mathcal{I}_t$
- **Martingale**: a process whose expected next value, given the past, equals its current value
  - With a constant required simple return m, the total-return price discounted at m, $P_t/(1+m)^t$, is a martingale Campbell, Lo & MacKinlay (1997); discounting at the risk-free rate gives a martingale only under the risk-neutral measure
  - P_t itself is one only if $m = 0$: a zero expected **simple** return, not a zero expected log return

The Jensen term: why averages differ

- The logarithm is concave: averaging log outcomes gives less than taking the log of their average, unless outcomes are constant.
- Equally likely simple returns of $+10\%$ and -10% have mean zero.
- The mean log return is $\frac{1}{2}\ln(1.1) + \frac{1}{2}\ln(0.9) = \frac{1}{2}\ln(0.99) \approx -0.00503$.
- Along a path with one rise and one fall, wealth 100 becomes $100(1.1)(0.9) = 99$: a zero arithmetic mean does not preserve compounded wealth.

Prices, log prices and information

- "$\mathbb{E}[P_{t+1} \mid \mathcal{I}_t] = P_t$" and "$\mathbb{E}[\ln(P_{t+1}/P_t) \mid \mathcal{I}_t] = \mu$" are different restrictions
  - They differ by discounting and by the Jensen term: $\mathbb{E}[\ln(1+R)] \approx \mathbb{E}[R] - \mathrm{Var}(R)/2$ for small returns ($\mathbb{E}[R]^2$ neglected)
  - $R = P_{t+1}/P_t - 1$: the simple return; $\ln(1+R)$: the log return; the mean log return is lower than the mean simple return by about half the variance
- Consequence: $\mathrm{corr}(\varepsilon_{t+1}, g(\mathcal{I}_t)) = 0$ for any function g of the past, $\varepsilon_{t+1} = r_{t+1} - \mu_t$
  - ε_{t+1}: the unexpected part of the return; in words, no past information is correlated with tomorrow's surprise
  - In particular, all autocorrelations $\rho_k = \mathrm{corr}(r_t, r_{t-k})$, $k \geq 1$, are zero when $\mu_t = \mu$; k: the lag in days

Rational bubble: definition

- Fundamental value: the present value of the cash flows expected from the asset.
  - For a share, these payments are dividends.
  - Discounting expresses future payments in today's money.
- A **rational bubble** is a positive price component above fundamental value.
  - It is sustained by the expectation of resale.
  - $P_t = P_t^* + B_t$: P_t is the price, P_t^* fundamental value, and B_t the bubble component.
- Rational expectations are consistent with the model, including crash risk.
  - They do not imply that a gain is certain.

Fundamental value, rational bubbles and fads

- With a constant required simple return m, the fair game is $\mathbb{E}[P_{t+1} + D_{t+1} \mid \mathcal{I}_t] = (1+m)\,P_t$; D_{t+1}: the cash flow (dividend) paid in $t+1$
  - Fundamental value $P_t^* = \mathbb{E}\big[\sum_{j\geq 1} D_{t+j}/(1+m)^j \mid \mathcal{I}_t\big]$: the present value of the expected cash flows at the required return; pricing error $B_t = P_t - P_t^*$
  - P_t^* satisfies the same recursion; subtracting: $P_t = P_t^* + B_t$ with $\mathbb{E}[B_{t+1} \mid \mathcal{I}_t] = (1+m)\,B_t$
- A fair game ties the price to the fundamental value only up to a **rational bubble**
  - A gap $B_t \neq 0$ is admissible only if it is expected to grow at the required rate m (Chapter 17)
  - A mean-reverting gap (fad) makes ε_{t+1} predictable and violates the fair game, although short-horizon tests have little power against it Summers (1986)
- The restriction is on the conditional mean only: the variance is not restricted, and volatility may be predictable (Chapter 1)

Rational bubble: the expected-growth condition

- m: the constant required simple return per period.
  - $\mathcal{I}_t$: information available at time t.
  - $\mathbb{E}[\cdot \mid \mathcal{I}_t]$: expectation conditional on that information.
- The bubble component must satisfy $\mathbb{E}[B_{t+1} \mid \mathcal{I}_t] = (1+m)B_t$, the recursion derived above from the fair-game condition.
  - In words: its expected growth rate equals the required return.
  - If a crash is possible, growth conditional on survival must be higher.
- A fad is a temporary price deviation expected to fade: a price gap of 20 expected to fall to 10 contrasts with a rational bubble expected to grow at the required rate.

Rational bubble: a numerical example

- A share pays 10 lei at the end of every year, forever.
  - The first dividend is paid one year from today.
  - The constant required return is $m = 10\%$; fundamental value is $P_t^* = 10/0.10 = 100$ lei.
- Today $P_t = 120$ lei, so $B_t = 20$ lei.
  - Without a crash, $B_{t+1} = 1.1 \times 20 = 22$ lei.
  - After next year's dividend, fundamental value remains 100 lei; the price is therefore 122 lei.
- The total simple return is $(122 + 10 - 120)/120 = 10\%$.

Rational bubble: what happens if it can burst?

- Initial bubble $B_t = 20$ lei and required return $m = 10\%$.
  - Survival probability $\pi = 0.8$; crash probability $1 - \pi = 0.2$.
  - A crash makes the bubble zero; investors know the probabilities.
- The growth condition requires an expected bubble of $(1 + m)B_t = 22$ lei.
  - Conditional on survival: $B_{t+1} = 22/0.8 = 27.5$ lei.
  - Check: $0.8 \times 27.5 + 0.2 \times 0 = 22$.
  - Conditional growth is $27.5/20 - 1 = 37.5\%$.
- Crash mechanism: Blanchard & Watson (1982), Bubbles, Rational Expectations and Financial Markets.
- The purchase price is 120 lei; the dividend at year-end is 10 lei.
- Survival: the ex-dividend price is 127.5 lei.
  - Simple return: $(127.5 + 10 - 120)/120 \approx 14.6\%$.
- Crash: the price falls to fundamental value, 100 lei.
  - Simple return: $(100 + 10 - 120)/120 \approx -8.3\%$.
- Expected return: $0.8 \times (17.5/120) + 0.2 \times (-10/120) = 10\%$.
  - The expected return satisfies fair game; either realized return can differ from it.

Rational bubble and the no-bubble condition

- Fair game restricts the expected return over the next period.
- The additional **no-bubble condition** is $\lim_{k\to\infty} \frac{\mathbb{E}[P_{t+k} \mid \mathcal{I}_t]}{(1+m)^k} = 0$.
  - k: the number of periods until resale.
  - The discounted expected resale price vanishes as the horizon grows.
  - Then the current price is supported entirely by cash flows.
- The numerical examples verify the one-period pricing recursion.
  - Existence in an economic equilibrium needs further assumptions: Tirole (1985), Asset Bubbles and Overlapping Generations.

The Grossman–Stiglitz Paradox

- Gathering information is costly (analysts, data, computing)
- If prices reflected **all** information, nobody would be paid for collecting it Grossman & Stiglitz (1980)
  - Then nobody would collect it, and prices could not reflect it
- Resolution: an **equilibrium degree of inefficiency**, just large enough to reward informed traders for their costs
- Markets with cheap information and low costs (large, liquid) should be closer to efficiency: a prediction for our comparison of developed, emerging and crypto markets
- Nor does efficiency rule out bubbles: it says they are hard to time (Chapter 17)

Random Walk Hypotheses RW1, RW2, RW3

- Log price $p_t = \ln P_t$ with $p_t = \mu + p_{t-1} + \varepsilon_t$; the three versions differ in what they assume about ε_t Campbell, Lo & MacKinlay (1997)
  - μ: the drift, the expected daily log return; ε_t: the shock of day t, with mean 0; so $r_t = p_t - p_{t-1} = \mu + \varepsilon_t$
- **RW1**: ε_t i.i.d., independent and identically distributed (the strongest; rules out volatility clustering)
  - Rejected by the stylised facts of Chapter 1
- **RW2**: ε_t independent but not identically distributed (variance may change)
- **RW3**: ε_t uncorrelated: $\mathrm{Cov}(\varepsilon_t, \varepsilon_{t-k}) = 0$, while $\mathrm{Cov}(\varepsilon_t^2, \varepsilon_{t-k}^2) \neq 0$ is allowed
  - The relevant hypothesis for real returns: tests must be robust to heteroskedasticity

Moments, tails and asymptotics

- Which assumptions make the test valid, and which alternatives can it detect?
- Prerequisite: from a pattern to a test
- Topics
- Contents

- The kth absolute moment is $\mathbb{E}[|r|^k]$
  - a finite fourth moment requires $\mathbb{E}[r^4] < \infty$, not merely finite observed returns.
- Heavy tails allow extreme values more often than a Normal benchmark
  - Student t with four degrees of freedom has finite variance but infinite fourth moment.
- An asymptotic approximation describes behaviour as sample size grows
  - it need not be accurate in a short sample.
- Mixing conditions require sufficiently distant blocks to become nearly independent at a suitable rate
  - adjacent daily returns need not be independent.

Autocorrelation and Its Robust Standard Error

- Sample autocorrelation: $\hat{\rho}_k = \sum_{t=k+1}^{T} e_t e_{t-k} / \sum_{t=1}^{T} e_t^2$, with $e_t = r_t - \bar{r}$
  - T: number of daily returns; $\bar{r}$: their mean; k: the lag; $\hat{\rho}_k > 0$: persistence, $\hat{\rho}_k < 0$: reversal
- Under RW1 (i.i.d.): $\mathrm{SE}(\hat{\rho}_k) \approx 1/\sqrt{T}$; 95% band $\pm 1.96/\sqrt{T}$
- Unpredictable mean with volatility clustering (finite fourth moments): $\mathrm{SE}(\hat{\rho}_k) \approx \sqrt{\hat{\tau}_k}$
  - $\hat{\tau}_k = \sum_t e_t^2 e_{t-k}^2 / \left(\sum_t e_t^2\right)^2$ (robust variance)
  - When large moves cluster, $\hat{\tau}_k > 1/T$: the i.i.d. band is too narrow and classical tests reject too often; every test below uses a variance that allows for this
- S&P 500: i.i.d. SE $= 0.012$, robust SE $= 0.027$, more than twice as large

Joint tests and reference distributions

- A portmanteau test combines several lag correlations into one statistic instead of selecting the most striking lag.
- A chi-squared distribution with m degrees of freedom is the distribution of a sum of squares of m independent standard Normal variables.
- For ten tested lags the reference is approximately χ^2_{10} under the stated assumptions
  - 25 exceeds the 5% critical value 18.31.
- A Wald test similarly tests coefficient restrictions jointly
  - five equal weekday means impose four restrictions, $\beta_1 = \beta_2$, $\beta_1 = \beta_3$, $\beta_1 = \beta_4$, $\beta_1 = \beta_5$, so the reference is χ^2_4, with 5% critical value 9.49.
- Five restrictions, all $\beta_d = 0$, would test a different null: every weekday mean equal to zero.

Portmanteau Tests: Ljung–Box and Its Robust Version

- Ljung & Box (1978): $Q(m) = T(T+2)\sum_{k=1}^{m} \hat{\rho}_k^2/(T-k) \sim \chi_m^2$ under i.i.d. returns
  - Tests jointly H_0: $\rho_1 = \cdots = \rho_m = 0$
  - m: the number of lags
  - χ_m^2: the chi-squared distribution with m degrees of freedom
  - A large Q rejects H_0
- Robust version Lobato, Nankervis & Savin (2001): $\tilde{Q}(m) = \sum_{k=1}^{m} \hat{\rho}_k^2/\hat{\tau}_k \to \chi_m^2$ under RW3 when the $\hat{\rho}_k$ are asymptotically uncorrelated (condition on the slide The Heteroskedasticity-Robust Statistic z^*)
- Three markets at $m = 10$ (5% critical value 18.31)
  - S&P 500: $Q = 96.3$ but $\tilde{Q} = 19.9$ ($p = 0.031$)
  - BET: $Q = 108.0$, $\tilde{Q} = 24.4$ ($p = 0.007$)
  - Bitcoin: $Q = 20.2$ ($p = 0.028$) but $\tilde{Q} = 11.9$ ($p = 0.294$)
- The robust test keeps the rejections for S&P 500 and BET, but not for Bitcoin

The Runs Test

- Nonparametric: uses only the **signs** of returns (zero returns dropped), so it is immune to heavy tails Wald & Wolfowitz (1940)
- A **run** is a maximal block of equal signs: ++|-|+++ has 3 runs; Seminar A5 counts them on 20 signs
- Moments of the number of runs R under independence, with n_1 positive and n_2 negative signs, $n = n_1 + n_2$
  - $\mathbb{E}[R] = \dfrac{2n_1n_2}{n} + 1, \qquad \mathrm{Var}[R] = \dfrac{2n_1n_2(2n_1n_2 - n)}{n^2(n-1)}, \qquad z = \dfrac{R - \mathbb{E}[R]}{\sqrt{\mathrm{Var}[R]}} \approx N(0,1)$
  - R: the observed number of runs; $\mathbb{E}[R]$, $\mathrm{Var}[R]$: its mean and variance if the signs were independent; $|z| > 1.96$ rejects at 5%
- Too **few** runs ($z < 0$): signs persist; too **many** ($z > 0$): sign reversals; the test assumes a constant probability of a positive day
- Three markets: S&P 500 $z = 4.34$ (more runs, as $\hat{\rho}_1 < 0$); BET $z = -6.05$ (fewer runs, as $\hat{\rho}_1 > 0$); Bitcoin $z = 2.68$ although $\hat{\rho}_1$ is not significant
  - signs and magnitudes carry different information; all markets: Appendix

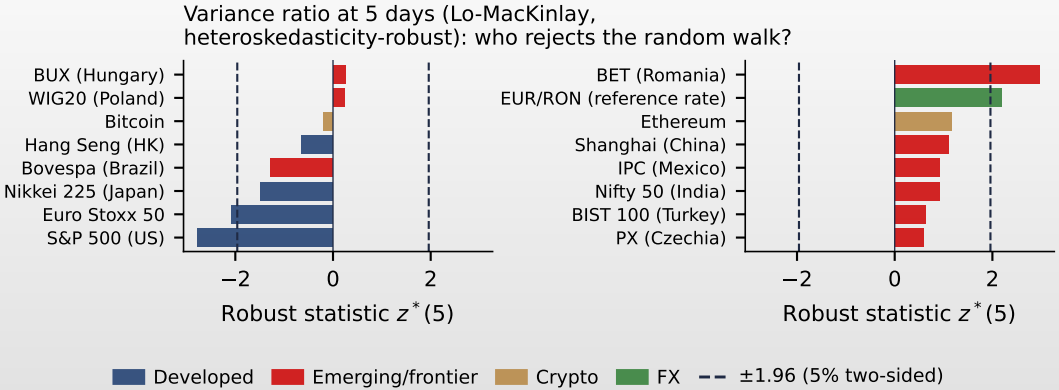
The Lo–MacKinlay Test

- Lo & MacKinlay (1988): overlapping q-day sums, with bias corrections
  - $\hat{\sigma}_a^2 = \frac{1}{T-1}\sum_t (r_t - \hat{\mu})^2, \quad \hat{\sigma}_c^2(q) = \frac{1}{m}\sum_{t=q}^{T}\left(\sum_{j=0}^{q-1} r_{t-j} - q\hat{\mu}\right)^2$
  - $m = q(T-q+1)(1-q/T); \quad \widehat{\mathrm{VR}}(q) = \hat{\sigma}_c^2(q)/\hat{\sigma}_a^2$
  - $\hat{\mu}$: mean daily return; $\hat{\sigma}_a^2$: variance of one-day returns; $\hat{\sigma}_c^2(q)$: variance of the overlapping q-day sums divided by q; m: a divisor that removes the small-sample bias
- Test statistic under RW1 (homoskedastic returns)
  - $z(q) = \dfrac{\widehat{\mathrm{VR}}(q) - 1}{\sqrt{2(2q-1)(q-1)/(3qT)}} \approx N(0,1)$
  - the denominator is the standard error of $\widehat{\mathrm{VR}}(q)$ under RW1; $|z| > 1.96$ rejects $\mathrm{VR}(q) = 1$ at 5%
- Homoskedastic: constant variance; this z over-rejects when volatility clusters (next slide)

The Heteroskedasticity-Robust Statistic z^*

- Under RW3 plus the conditions of Lo & MacKinlay (1988) (finite moments, weak dependence (mixing), $\mathbb{E}[\varepsilon_t^2 \varepsilon_{t-j}\varepsilon_{t-k}] = 0$ for $j \neq k$), the $\hat{\rho}_k$ are asymptotically uncorrelated with variances $\hat{\delta}_k / T$
  - $\hat{\delta}_k = T\hat{\tau}_k$, with $\hat{\tau}_k$ of the autocorrelation slide: the robust variance of $\sqrt{T}\hat{\rho}_k$, equal to 1 without volatility clustering
- Asymptotic variance of $\widehat{\mathrm{VR}}(q)$ and the test statistic
  - $\hat{\theta}(q) = \sum_{k=1}^{q-1} \left(\frac{2(q-k)}{q} \right)^2 \hat{\delta}_k, \qquad z^*(q) = \frac{\sqrt{T}\,(\widehat{\mathrm{VR}}(q) - 1)}{\sqrt{\hat{\theta}(q)}} \approx N(0,1)$
  - $2(q-k)/q$: the weight of $\hat{\rho}_k$ in $\widehat{\mathrm{VR}}(q)$; $|z^*| > 1.96$ rejects $\mathrm{VR}(q) = 1$ at 5%
- RW3 alone is not enough: the conditions hold for GARCH with symmetric innovations, but can fail with a leverage effect
  - Without the last condition, $\hat{\theta}(q)$ needs the full covariance matrix of the $\hat{\rho}_k$
- S&P 500, $q = 2$: $\widehat{\mathrm{VR}} = 0.901$, $z = -8.12$ but $z^* = -3.64$: still significant, but half as large in absolute value (Seminar B1 works it step by step)

Several Horizons and Markets: the Chow–Denning Test (1/2)



MFM_ch2_variance_ratio

Several Horizons and Markets: the Chow–Denning Test (2/2)

- Bars: robust $z^*(5)$ for all 16 series; dashed lines ± 1.96
- Single horizon, reject at 5%: BET (2.98), EUR/RON (2.19), S&P 500 (-2.77), Euro Stoxx 50 (-2.08)
- Four horizons at 5% each: false-rejection probability up to $1 - 0.95^4 = 18.5\%$
  - Chow & Denning (1993): CD $= \max_i |z^*(q_i)|$, the largest of the four $|z^*|$, over $q_i = 2, 5, 10, 20$; 5% critical value 2.49, above 1.96 to pay for the four looks
- The joint test rejects RW3 for 5 of the 16 series: BET, S&P 500, IPC, EUR/RON, Ethereum; Bitcoin $p = 0.835$
- No ranking "developed = efficient": the direction differs (reversal vs persistence)

MFM_ch2_variance_ratio

What nonlinear tests examine

- **BDS**: Brock–Dechert–Scheinkman–LeBaron test.
- **MDH**: Martingale Difference Hypothesis.
- BDS compares the frequency of nearby return sequences with the frequency implied by i.i.d. observations
  - excess clustering can signal dependence. Brock, Dechert, Scheinkman & LeBaron (1996)
- For example, persistent quiet and volatile regimes can trigger BDS even if tomorrow's conditional mean is zero.
- The generalised spectral test looks for mean dependence on nonlinear transformations of past returns
  - it can detect dependence missed by ordinary correlations. Escanciano & Velasco (2006)
- Automatic portmanteau chooses the number of tested lags using a penalty
  - it can reject MDH through linear dependence but does not detect every nonlinear alternative. Escanciano & Lobato (2009)

Beyond Linear Predictability: the Martingale Difference Hypothesis

- **MDH**: $\mathbb{E}[r_t \mid \mathcal{I}_{t-1}] = \mu$: no function of the past predicts the mean, linear or not
- RW3 (all $\rho_k = 0$) is weaker. Counterexample: $r_t = \varepsilon_t + \theta(\varepsilon_{t-1}^2 - \sigma^2)$, ε_t i.i.d. symmetric
  - $\sigma^2 = \mathrm{Var}(\varepsilon_t)$, so the extra term has mean 0; $\theta \neq 0$: how strongly yesterday's squared shock moves today's mean
  - all $\rho_k = 0$, yet $\mathbb{E}[r_t \mid \varepsilon_{t-1}] = \theta(\varepsilon_{t-1}^2 - \sigma^2)$ is predictable: autocorrelation tests cannot see it
  - Tests of the MDH itself: the generalised spectral test Escanciano & Velasco (2006); an automatic portmanteau Escanciano & Lobato (2009); details and results: Appendix
- **BDS** Brock, Dechert, Scheinkman & LeBaron (1996) tests i.i.d., not the MDH: GARCH alone makes it reject
  - A BDS rejection is evidence against RW1, not against weak-form efficiency

Predetermined variables, bias and consistency

- Why does a persistent predictor distort inference in a finite sample?
- Prerequisite: forecasts on new data
- Topics
- Contents

- A predetermined predictor is known before the return shock it predicts
  - strict exogeneity additionally rules out correlation with shocks at every date.
- Today's dividend–price ratio can forecast tomorrow, yet tomorrow's price shock changes tomorrow's ratio: this feedback obstructs strict exogeneity.
- Bias is the estimator's repeated-sample mean minus the true parameter: mean slope 0.03 when the true slope is zero means bias 0.03.
- Consistency means convergence to the true parameter as data grow
  - it does not imply zero bias in a finite sample.

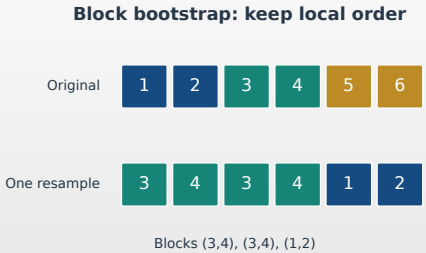
HAC standard errors and bandwidth

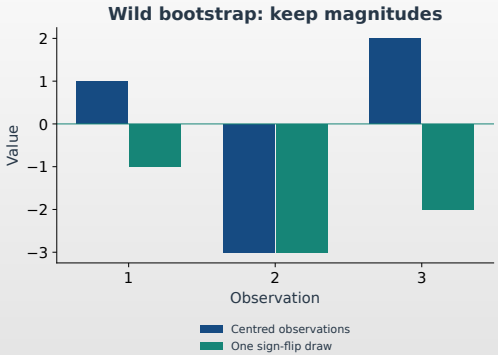
- HAC means heteroskedasticity and autocorrelation consistent: standard errors allow both changing variance and serial dependence under regularity conditions.
- Newey–West estimates long-run variance by adding weighted lag covariances to the variance
  - it changes uncertainty estimates, not the OLS coefficients.
- For a stationary series with variance 1, lag-1 covariance 0.2 and all other covariances zero, long-run variance is $1 + 2(0.2) = 1.4$.
- Bandwidth specifies how many lags enter the estimate
  - bandwidth 5 uses lags 1–5, with weights depending on the estimator.

Bootstrap: constructing repeated samples

- Bootstrap approximates the sampling distribution by generating many artificial samples and recomputing the statistic.
- An i.i.d. bootstrap draws observations with replacement: from $(1, 2, 3)$ one draw can be $(2, 2, 1)$
  - serial data need a dependence-aware design.
- Block bootstrap resamples consecutive blocks
  - drawing blocks $(1, 2)$ and $(3, 4)$ preserves within-block order, unlike shuffling individual days.
- Wild bootstrap multiplies centred returns by random signs: $(1, -3, 2)$ may become $(-1, -3, -2)$, keeping absolute magnitudes
  - validity depends on the null model.

What each bootstrap preserves





Blocks preserve local ordering; sign flips preserve absolute magnitudes. They impose different assumptions on the artificial samples.

Predictive Regressions and the Stambaugh Bias (1/2)

- Question: does a valuation ratio known today forecast next period's return?
- Model: $r_{t+1} = \alpha + \beta x_t + u_{t+1}, \quad x_{t+1} = \theta + \rho x_t + v_{t+1}$, with $\mathrm{Cov}(u, v) = \sigma_{uv}$
  - r: excess return; x: log dividend–price ratio $dp_t = \ln D_t - \ln P_t$, persistent ($\rho \approx 1$)
  - D_t: dividends paid over the last year; β: the predictive slope, 0 if returns are unpredictable; α, θ: intercepts
  - ρ: the persistence of x; u, v: the shocks of the two equations; σ_{uv}: their covariance; $\sigma_v^2 = \mathrm{Var}(v)$; T: number of periods

Bonferroni Q: allowing for uncertain persistence

- Campbell–Yogo inference allows for uncertainty about predictor persistence instead of treating its estimate as known. Campbell & Yogo (2006)
- It combines a confidence set for persistence with tests of the predictive slope over that set, allocating the error probability across the steps.
- For example, if plausible persistence values range from 0.95 to 1, a conclusion must account for that range, not only the point estimate 0.98.
- This Q-test concerns predictive regressions
  - it is not the Ljung–Box Q statistic for autocorrelations.

Predictive Regressions and the Stambaugh Bias (2/2)

- x_t is predetermined, not strictly exogenous: OLS is consistent but biased Stambaugh (1999)
  - Project u on v: $u = (\sigma_{uv}/\sigma_v^2)\, v + e$, so $\hat{\beta} - \beta = (\sigma_{uv}/\sigma_v^2)(\hat{\rho} - \rho) + \text{noise}$; e: the part of u unrelated to v
  - The step uses the same months and the same demeaned x_t in both regressions; the noise term has mean zero when e is independent of the whole path of x, e.g. (u, v) jointly Normal and i.i.d.
  - $\hat{\rho}$ is biased downward in small samples Kendall (1954), and this error passes into $\hat{\beta}$
  - With $\mathbb{E}[\hat{\rho} - \rho] \approx -(1 + 3\rho)/T$: $\quad \mathbb{E}[\hat{\beta} - \beta] \approx -\dfrac{\sigma_{uv}}{\sigma_v^2}\,\dfrac{1 + 3\rho}{T}$
- Price up $\Rightarrow$ dp down: $\sigma_{uv} < 0$, so the bias is **upward**: predictability looks stronger than it is
- Near a unit root the t-statistic is not $N(0, 1)$ either: valid inference by a Bonferroni Q-test over a confidence interval for ρ Campbell & Yogo (2006)

Worked Example: the Stambaugh Bias for the S&P 500

- Monthly log excess return regressed on the lagged dp_t, 1934–2026, $T = 1\,111$ months; data construction: Appendix
- OLS: $\hat{\beta} = 0.267$ (% per month per unit of dp), Newey–West $t = 0.91$: not significant even before any correction
- Inputs: $\hat{\rho} = 0.9988$, $\text{corr}(\hat{u}, \hat{v}) = -0.97$, $\hat{\phi} = \hat{\sigma}_{uv}/\hat{\sigma}_v^2 = -0.925$ (as fractions, not in %)
- Substitution: bias $= -100\,\hat{\phi}\,(1 + 3\hat{\rho})/T = 0.333$ (in %, like $\hat{\beta}$)
  - corrected slope $= 0.267 - 0.333 = -0.066$: the whole estimate is bias
  - the correction is itself estimated: formal inference by Campbell & Yogo (2006) or a bootstrap
  - Kendall's formula is first order in $1/T$ and least accurate when $T(1 - \hat{\rho})$ is small, here 1.4 using the unrounded persistence estimate: read -0.066 as indicative of the sign and size, not as a precise slope
- Simulation with the same ρ and $\text{corr}(u, v)$, $\beta = 0$, $T = 360$ months: the nominal 5% t-test rejects $\beta = 0$ in 26.5% of samples

Comparing forecasts and economic value

- Diebold–Mariano tests the mean loss difference between forecasts
  - Clark–West adjusts squared losses for the extra estimation noise of a nested model.
- A one-sided test here asks whether the adjusted gain is positive
  - a two-sided test would also count a sufficiently negative gain as a departure from zero.
- Sharpe ratio divides mean excess return by its SD at the same frequency: monthly mean 1%, SD 5%, Sharpe $= 0.2$.
- A mean–variance investor trades off expected return against variance: at equal expected return, the lower-variance portfolio is preferred.

Out-of-Sample Predictability: R^2_{OS} and the Clark–West Test

- Recursive forecasts: at each month t, re-estimate on data up to $t-1$ only (expanding window) and forecast $\hat{r}_t$
  - benchmark: the historical mean $\bar{r}_t$ of the same data; $R^2_{OS} = 1 - \sum_t (r_t - \hat{r}_t)^2 / \sum_t (r_t - \bar{r}_t)^2$; $R^2_{OS} > 0$: the predictor beats the mean
  - Most predictors have $R^2_{OS} < 0$ Welch & Goyal (2008); the 2022 update confirms it Goyal, Welch & Zafirov (2024)
  - Pseudo out-of-sample, as in Welch & Goyal (2008): monthly dividends are interpolated from quarterly totals, so dp_t is not strictly real-time
- Nested models: under H_0 the larger model estimates a zero β, which inflates its MSPE (mean squared prediction error); the Diebold–Mariano test Diebold & Mariano (1995) then rejects less often than its nominal level
  - Clark & West (2007): $f_t = (r_t - \bar{r}_t)^2 - \left[(r_t - \hat{r}_t)^2 - (\bar{r}_t - \hat{r}_t)^2\right]$; one-sided t on the mean of f_t
  - f_t: the squared-error gain of the predictor, plus the term $(\bar{r}_t - \hat{r}_t)^2$ that removes the noise of estimating β; a positive mean favours the predictor
- Campbell & Thompson (2008): impose the sign of β and a non-negative forecast

The Dividend–Price Ratio: Estimation and Interpretation

- Cumulative squared-error gain over the historical mean, forecasts from 1970; rising line: *dp* helps
- 680 forecasts: $R^2_{OS} = -0.51\%$; the Campbell–Thompson restrictions bind in 5.7% of months and leave R^2_{OS} practically unchanged, at -0.51%
- Clark–West $t = 0.42$ (one-sided $p = 0.34$): no significant out-of-sample predictability
- The gain peaks in 1990 and is lost in the 1990s boom, when *dp* fell to record lows; after 2010 it erodes again
- Economic relevance: with monthly Sharpe ratio 0.160, even $R^2 = 0.5\%$ would raise by 20% the expected return of a mean–variance investor who times the market with the forecast; formula:

▸ Appendix

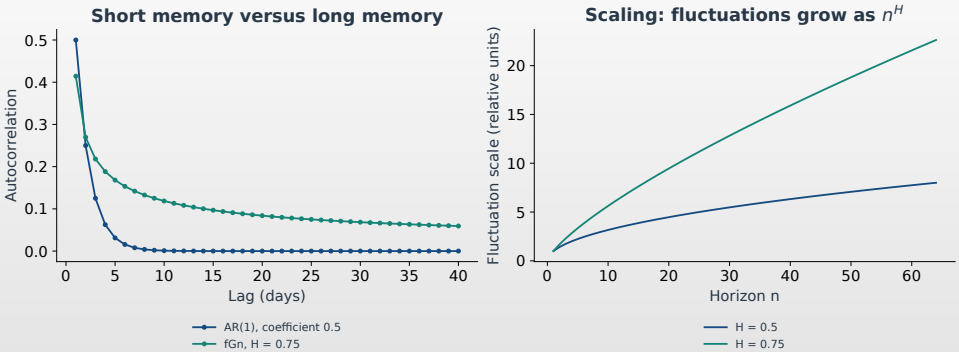
MFM_ch2_predictability

Short memory, long memory and scaling

- Does dependence persist across horizons and market regimes?
- Prerequisite: from a pattern to a test
- Topics
- Contents

- Short memory has absolutely summable autocovariances
  - an AR(1) with coefficient 0.5 has correlations 0.5^k at lag k.
- Long memory here means positive dependence decaying so slowly that the autocovariance sum diverges
  - power-law decay can produce this.
- Scaling $F(n) \propto n^H$ means multiplying the horizon by 2 multiplies fluctuations by 2^H: about 1.41 at $H = 0.5$, or 1.68 at $H = 0.75$.
- Antipersistence describes offsetting increments, as in fractional Gaussian noise (fGn) with $H < 0.5$
  - fGn is a stationary Gaussian increment model with partial-sum variance proportional to n^{2H}.

Dependence across lags and horizons

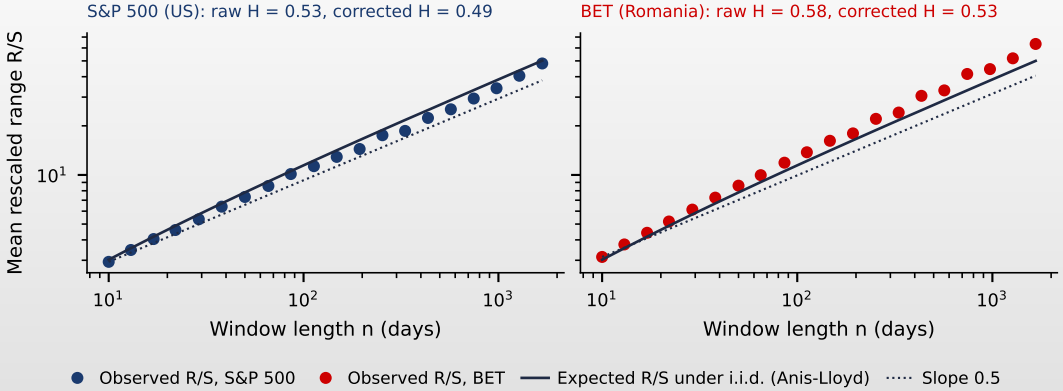


AR(1) correlations fade geometrically; the fGn example fades slowly. The right panel compares fluctuation growth across horizons.

The Rescaled Range (R/S) Statistic

- Using the rescaled range, Hurst (1951) found that wet years on the Nile followed wet years more often than chance allows; history: Appendix
- Construction for a window of n returns with mean $\bar{r}_n$ and standard deviation s_n
  - Cumulative deviations $Y_k = \sum_{i=1}^{k}(r_i - \bar{r}_n)$, $k = 1, \ldots, n$
  - Range $R_n = \max_k Y_k - \min_k Y_k$; rescaled range R_n/s_n
- Scaling law: $\mathbb{E}[R_n/s_n] \approx c\, n^H$, where H is the **Hurst exponent**
  - c: a constant; taking logs, $\ln(R/S) \approx \ln c + H \ln n$, so H is a slope
  - Estimate: slope of $\ln(R/S)$ on $\ln n$, averaging R/S over non-overlapping windows
- $H = 0.5$: no long memory (short-run dependence allowed); $H > 0.5$: persistence; $H < 0.5$: anti-persistence
  - **Long memory** ($H > 0.5$): autocorrelations decay so slowly that their sum diverges, $\rho_k \sim k^{2H-2}$

R/S Analysis: S&P 500 and BET (1/2)



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R/S Analysis: S&P 500 and BET (2/2)

- Dots: average R/S by window length n; solid line: the Anis & Lloyd (1976) expectation under i.i.d. returns from the Normal distribution; exact formula: Appendix
- Small-sample bias: even for i.i.d. returns the raw slope exceeds 0.5; the Anis–Lloyd corrected exponent $H_{AL} = 0.5+$ slope of $\ln(R/S) - \ln \mathbb{E}[R_n/s_n]$
- S&P 500: raw 0.530, corrected 0.486: the "persistence" was the artefact
- BET: raw 0.578, corrected 0.533: persistence remains
- Open: is BET's $H > 0.5$ **long** memory or just $\hat{\rho}_1 = 0.109$? Classical R/S cannot tell

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Hurst Exponent and Variance Ratio: One Horizon Is Not Enough

- For exactly self-similar increments (fractional Gaussian noise): $\mathrm{Var}(r_t + \cdots + r_{t-q+1}) = q^{2H}\mathrm{Var}(r_t)$
- Hence $\mathrm{VR}(q) = q^{2H-1}$ and the **implied** $H(q) = 0.5 + \ln \mathrm{VR}(q)/(2 \ln q)$: BET 0.553, S&P 500 0.453 at $q = 20$
- Short-memory counterexample: an AR(1) with $\rho_1 = -0.1$, $\rho_k = (-0.1)^k$
  - $\mathrm{VR}(20) = 0.826$, so $H(20) = 0.468$: "anti-persistent" without any long memory
  - yet $H(250) = 0.482$: for short memory $H(q) \to 0.5$ as $q \to \infty$ (Seminar A2–A3)
  - an AR(1) with BET's $\hat{\rho}_1 = 0.109$ gives $H(20) = 0.535$: most of the gap between BET's $H(20)$ and 0.5 is short memory
- Separating the two needs a statistic built for it: Lo's modified R/S (next slide)

Lo's Modified R/S Test

- Lo (1991): short-run autocorrelation inflates R/S and mimics long memory
- Modification: s_n is replaced by a long-run (HAC) standard deviation $\hat{\sigma}_\ell$ with ℓ lags
  - $\hat{\sigma}_\ell^2 = \hat{\gamma}_0 + 2\sum_{j=1}^{\ell}\left(1 - \frac{j}{\ell+1}\right)\hat{\gamma}_j$, $\hat{\gamma}_j$: autocovariance at lag j ($\hat{\gamma}_0$: the variance), with declining weights
  - $V = \dfrac{\max_k Y_k - \min_k Y_k}{\hat{\sigma}_\ell \sqrt{T}}$; Y_k: cumulative deviations over the full sample of T returns; bandwidth ℓ chosen as in Andrews (1991)
- Under short memory, V lies in $[0.809, 1.862]$ with probability 95%
- Full samples: S&P 500 $V = 1.559$, Bitcoin 1.547, PX 1.750, EUR/RON 1.429: short memory is not rejected
  - BET $V = 1.961$ ($\ell = 7$) and Ethereum $V = 1.980$: just above 1.862, weak evidence of long memory

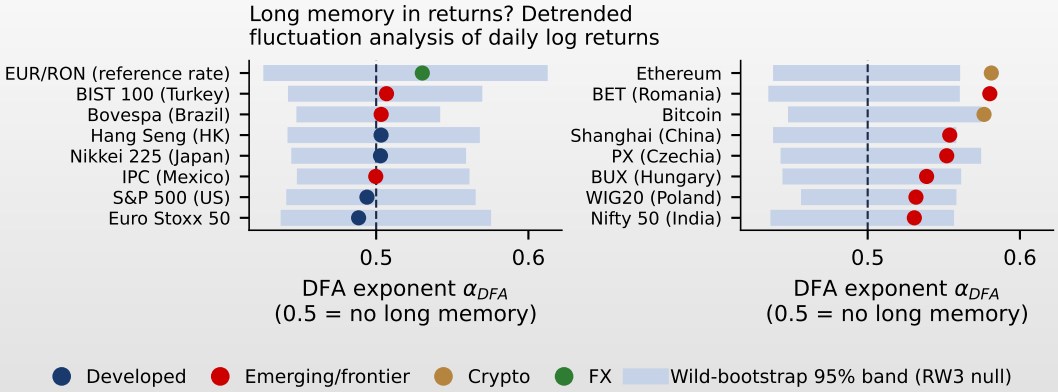
Detrended Fluctuation Analysis (DFA)

- Peng et al. (1994): linear detrending of the profile removes a locally constant mean of returns; trends in returns need higher-order DFA
- Algorithm in three steps (Seminar B5 shows one window step by step)
  - Profile y_k: the cumulative deviations Y_k of R/S, over the full sample; split into boxes of length n
  - In each box fit a linear trend and keep the residuals
  - $F(n)$ = root mean square of all residuals; repeat for many n
- $F(n) \propto n^{\alpha_{DFA}}$: the DFA exponent α_{DFA} plays the role of H ($\alpha_{DFA} = 0.5$: no long memory)
- Null band by **wild bootstrap**: $e_t\eta_t$, $\eta_t = \pm 1$ with probability $1/2$ (199 replications); η_t: a random sign
  - Keeps the volatility path $|e_t|$ (allowed under RW3) and destroys serial correlation; shuffling would destroy both
  - Implicit null: signs symmetric and independent given $|e_t|$, stronger than RW3: a benchmark, not an exact RW3 test and not a confidence interval for α_{DFA}

Frequency, periodogram and local Whittle

- A frequency describes cycles per unit time: a ten-day cycle has frequency $1/10$ per day
  - low frequencies describe slow variation.
- Spectral density allocates variance across frequencies
  - the periodogram is its raw sample analogue, built from squared Fourier amplitudes.
- Fourier components are sine and cosine waves: combining their amplitudes represents oscillations at different frequencies.
- Local Whittle fits the low-frequency spectral slope to estimate d
  - $d = 0.1$ corresponds to $H = 0.6$ in the stated scaling model.
  - Its bandwidth m counts frequencies, not lags. Robinson (1995)

DFA Exponent Across Markets (1/2)



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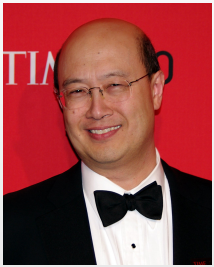
DFA Exponent Across Markets (2/2)

CI: Confidence Interval.

- Dots: α_{DFA}; bars: sign-flip 95% null band (volatility kept, no autocorrelation)
- Inside the band: S&P 500 0.494, Euro Stoxx 50 0.488, Nikkei 0.503, PX 0.552, Shanghai 0.554
- Above (3 of 16): Ethereum 0.581, BET 0.580 (band to 0.561), Bitcoin 0.576, borderline (band to 0.575)
- A shuffled band flags 5: PX and Shanghai were "memory" only through volatility clustering
- Cross-checks: Lo's V rejects short memory marginally for BET and Ethereum, not for Bitcoin; the local Whittle estimator of the long-memory parameter $d = H - 0.5$ for BET: $\hat{d} = 0.118$, 95% CI $[0.062, 0.174]$; details: Appendix
- Lesson: memory in returns is weak and fragile; memory in volatility (Chapter 1) is strong

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Adaptive Markets Hypothesis (AMH)



Andrew W. Lo, MIT

Photo: David Shankbone (2012); CC BY 3.0; Wikimedia Commons

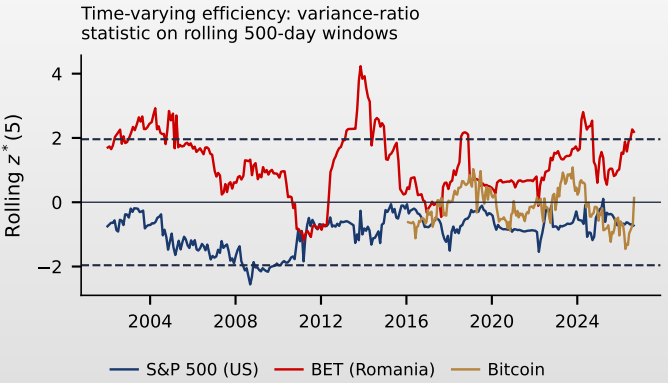
- Lo (2004); book-length version Lo (2017)
- Markets are an ecology of competing species (traders, funds, algorithms) that adapt by trial and error, not by perfect optimisation
  - efficiency is a matter of degree and depends on the number of competitors, the profit opportunities and the environment
- Prediction: efficiency **varies over time and with market conditions**; anomalies appear, weaken and may reappear
- Caution: time-varying statistics alone do not identify the AMH, since an Efficient Market Hypothesis (EMH) with time-varying risk premia also allows them; comparison table: ▸ Appendix
- Evidence of time variation: survey Lim & Brooks (2011); long-run indices Urquhart & McGroarty (2016)

Measuring Time-Varying Efficiency

- Compute a statistic on a **rolling window**: 500 trading days (about two years), moved forward by 21 days (one month)
  - Statistics: $z^*(5)$ of the variance-ratio test and the DFA exponent
- Null band for DFA: wild bootstrap **inside each window** (199 sign flips), so the band follows the window's volatility
  - An i.i.d. Student-t_4 band $[0.378, 0.614]$ ignores volatility clustering (allowed under RW3), which raises α_{DFA}
- Two cautions for reading the results
  - Overlapping windows are strongly correlated: runs of rejections are not independent evidence
  - Even under the null about 5% of windows reject; a share near 5% is a description, not a calibrated test of stability

Rolling Variance Ratio: S&P 500, BET, Bitcoin

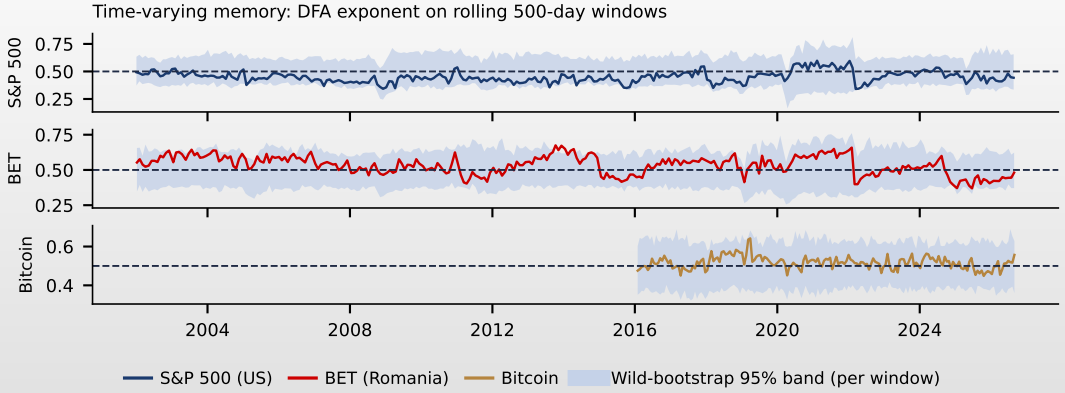
FTSE: Financial Times Stock Exchange (FTSE Russell, index provider).



- $z^*(5)$ on rolling 500-day windows; dashed lines ± 1.96
- Share of windows with $|z^*(5)| > 1.96$: S&P 500 6.4%, BET 22.4%, Bitcoin 0.0%
- S&P 500: the full-sample rejection sits in a few turbulent windows
- BET: rejections cluster in episodes, then fade; FTSE upgrade to Secondary Emerging effective 21 Sep 2020, a candidate change in the investor base; Seminar C2 and ▸ Appendix
- Bitcoin, 2014–2026: no rejection in any window

MFM_ch2_rolling_efficiency

Rolling DFA Exponent: S&P 500, BET, Bitcoin (1/2)



MFM_ch2_rolling_efficiency

Rolling DFA Exponent: S&P 500, BET, Bitcoin (2/2)

- α_{DFA} on rolling 500-day windows; shaded: wild-bootstrap 95% band computed in each window
- Outside their own band: S&P 500 2.0%, BET 2.7%, Bitcoin 0.0% of windows
- With the fixed i.i.d. Student-t_4 band: 6.4%, 12.9%, 1.1%: the apparent BET "episodes" were volatility clustering
- Low power: the BET band averages $[0.36, 0.64]$; 500 observations cannot pin down a scaling exponent

MFM_ch2_rolling_efficiency

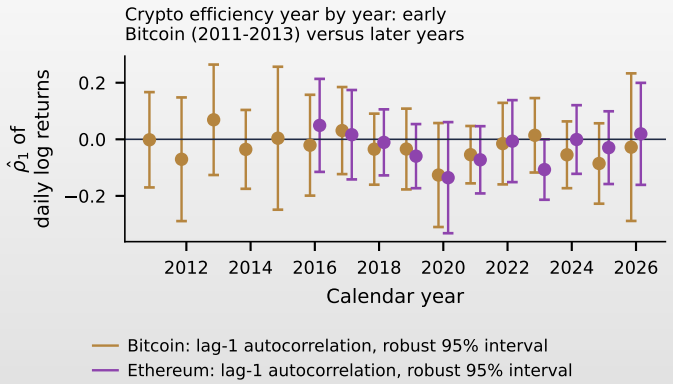
Indicators and controls

- A dummy equals 1 when a condition holds and 0 otherwise: a recession dummy marks recession months.
- In a regression, a control variable holds an observed characteristic fixed statistically
  - including volatility compares periods with similar volatility.
- A confounder affects both the putative cause and the outcome: a crisis may change both trading participation and predictability.
- An association after controls is not automatically causal: omitted confounders can remain.

The AMH, Tested

- Kim, Shamsuddin & Lim (2011): century-long US data; predictability varies with market conditions (crises, bubbles), not along a trend
  - This turns the AMH narrative into a regression with testable coefficients
- Our version: regress the rolling $z^*(5)$ on the window's annualised volatility (%) and a crisis dummy (window overlaps Sep 2008–Jun 2009 or Feb–Jun 2020)
  - Newey–West errors with 24 lags: consecutive windows share 95% of their days
- S&P 500: crisis -0.67 ($t = -2.37$), volatility 0.006 ($t = 0.27$), $R^2 = 0.20$
- BET: crisis -0.82 ($t = -2.77$), volatility -0.008 ($t = -0.36$), $R^2 = 0.15$
- Crisis windows are associated with lower $z^*(5)$ (towards reversal) in both markets, beyond volatility: a conditional association, not a causal effect
  - z^* mixes $\widehat{\mathrm{VR}} - 1$ and its standard error; state dependence is consistent with the AMH but does not identify it uniquely

Crypto Efficiency Year by Year



- Lag-1 autocorrelation for each calendar year, robust 95% intervals; Bitcoin from 2011
- Only 1 of 27 yearly $\hat{\rho}_1$ has $|t| > 1.96$ (Ethereum 2023, -0.107), against $0.05 \times 27 = 1.35$ expected by chance
- Unlike Urquhart (2016), who rejects efficiency in the early years: other tests (including BDS), another price source and classical errors; literature:
  ▸ Appendix
- Power: one year of 365 returns gives a robust SE of about 0.06–0.13, so small effects are invisible

MFM_ch2_crypto_efficiency

SD, SE and uncertainty

- Standard deviation (SD) measures dispersion of observations
  - standard error (SE) measures dispersion of an estimator across repeated samples.
- For 100 i.i.d. returns with SD 2%, the sample mean has SE $2\%/\sqrt{100} = 0.2\%$.
  - Dependence changes this calculation.
- A 95% confidence interval is a procedure covering the fixed unknown parameter in 95% of repeated samples, under its assumptions.
- A null band instead shows typical estimates if a specified null is true: a simulated band around $H = 0.5$ is not a confidence interval around an estimated H.

Cross-sectional dependence and standardisation

- How do dependence and event-induced variance affect the test?
- Prerequisite: prices and news
- Topics
- Contents

- A time series follows one asset across dates
  - a cross-section compares many assets at one date.
- Two banks hit by the same news need not be independent observations: their abnormal returns can move together.
- Standardisation divides a centred quantity by its SD: an abnormal return of -4% with SD 2% is -2 standard deviations.
- Event-induced variance means the event itself raises dispersion
  - using only quiet pre-event days can understate uncertainty.

Event Studies: Inference Done Properly

- Prediction-error variance: $\mathrm{Var}(AR_\tau) = \sigma_\varepsilon^2\left[1 + \frac{1}{T_0} + \frac{(r_{m,\tau} - \hat{\mu}_m)^2}{(T_0 - 1)\hat{\sigma}_m^2}\right]$, T_0 = estimation days MacKinlay (1997)
  - $\hat{\mu}_m$, $\hat{\sigma}_m^2$: mean and variance of the market return in the estimation window; the bracket grows when the event-day market move is unusual
  - Bank tax, BET $r_{m,0} = -11.9\%$: the bracket is 2.12, so TLV's t shrinks in absolute value from -10.0 to -6.85; BRD: $AR_0 = -5.6\%$, $t = -4.03$
- Several firms: standardise each AR by its own prediction error and use the cross-sectional variance, robust to event-induced variance Boehmer, Musumeci & Poulsen (1991)
- Same event day for all firms: abnormal returns are cross-correlated and cross-sectional tests over-reject; the equal-weighted portfolio test already contains the correlation
  - TLV and BRD: $\bar{\varrho} = 0.153$; mean AR_0 with an independence $t = -7.55$, portfolio $t = -7.03$
  - $\bar{\varrho}$: the average correlation between the firms' abnormal returns; N: number of firms
  - Kolari & Pynnönen (2010) correct a cross-sectional test by $\sqrt{(1 - \bar{\varrho})/(1 + (N - 1)\bar{\varrho})}$; with ten such firms, $(1 + 9\bar{\varrho})/(1 - \bar{\varrho}) = 2.81$: the test would understate the variance of the mean AR 2.81 times

Size, value, momentum and data snooping

- Do calendar patterns and behavioural stories survive risk and data-snooping checks?
- Prerequisite: conclusions that survive scrutiny
- Topics
- Contents

- Size compares small and large market capitalisations
  - value compares cheap stocks relative to fundamentals, such as book equity, with expensive stocks.
- Cross-sectional momentum ranks assets: if A gained 10% and B gained 2%, a relative strategy may buy A and short B.
- Time-series momentum uses each asset's own past sign: both A and B would receive positive signals in that example.
- Data snooping selects a rule after trying many on the same data
  - reporting only the best weekday out of many searches hides the chance of a lucky result.

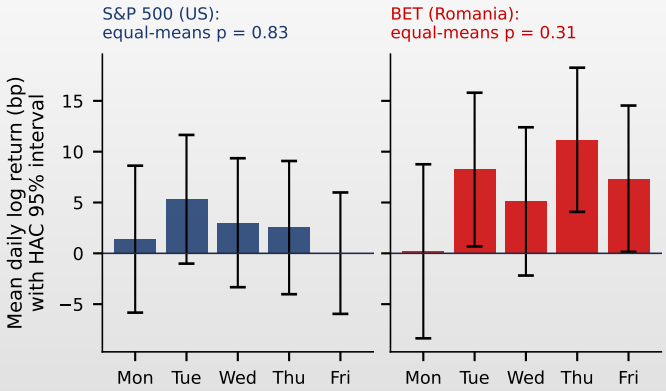
ETFs, placebo and price discovery

- A spot ETF is an exchange-traded fund holding the underlying asset
  - buying one unit gives exposure through the fund rather than a futures contract.
- A futures contract fixes today the terms for a future trade
  - buy-and-hold means retaining a purchased position rather than changing its sign with a signal.
- A placebo repeats the analysis at a date without the event: a similar apparent change in 2019 weakens a claim that the 2024 launch caused the change.
- Price discovery is the incorporation of information into prices: if exchange A reacts to news first and B follows, A may contribute more to price discovery.

Anomalies and Calendar Effects

- An **anomaly** is a return pattern the asset-pricing model cannot explain: an omitted risk factor (misspecified model), mispricing (behavioural bias) or data mining
  - Families: calendar, time-series (index momentum), cross-sectional (size, value, momentum: **Chapter 3**)
  - Returns 26% lower out of sample and 58% lower after publication McLean & Pontiff (2016)
- Calendar effects: Monday French (1980); January Rozeff & Kinney (1976); turn of the month (TOM: last trading day and first three of the next month) Lakonishok & Smidt (1988); monthly effect (first half of the month) Ariel (1987)
  - Day of week: $r_t = \sum_{d=1}^{5} \beta_d D_{d,t} + u_t$, Wald test of $\beta_1 = \cdots = \beta_5$ (four equality restrictions, reference χ^2_4); TOM and January: one dummy each
  - $D_{d,t} = 1$ if day t is weekday d (Monday, ..., Friday), else 0; β_d: the mean return on weekday d; u_t: the residual
  - Newey–West HAC errors Newey & West (1987), 5 daily or 3 monthly lags: residuals are heteroskedastic and autocorrelated

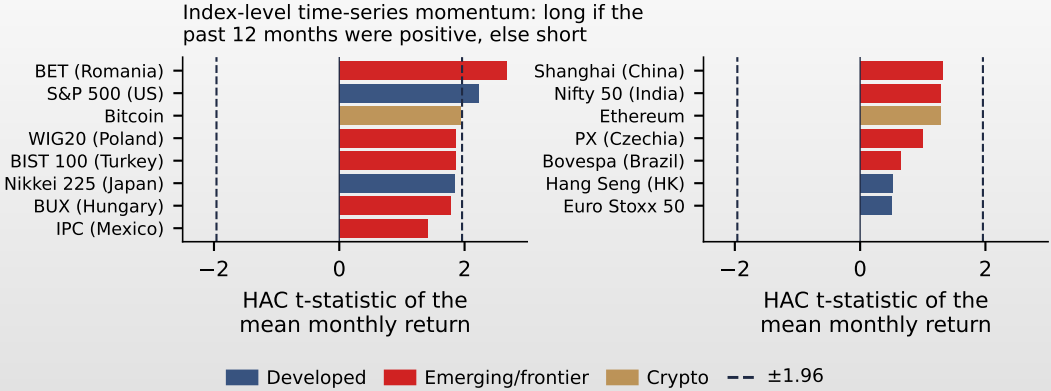
Day-of-the-Week Effect: S&P 500 and BET



- Mean daily log return by weekday, in basis points (bp; 1 bp = 0.01%), 2000–2026, HAC 95% intervals
- S&P 500: from 0.0 bp (Friday) to 5.3 bp (Tuesday); equal-means $p = 0.83$: no Monday effect since 2000, unlike French (1980)
- BET: Thursday 11.2 bp ($t = 3.09$), yet the joint test gives $p = 0.31$
- Two different nulls: a day's mean equal to zero (mixes the day effect with the positive average return) vs all five means equal (the day effect itself)

MFM_ch2_calendar_anomalies

Momentum and Index-Level Time-Series Momentum (1/2)



MFM_ch2_tsmom

Momentum and Index-Level Time-Series Momentum (2/2)

- Reversal of 3–5-year losers De Bondt & Thaler (1985); cross-sectional momentum over 3–12 months Jegadeesh & Titman (1993) (Chapter 3)
- TSMOM Moskowitz, Ooi & Pedersen (2012): signal $s_t = \text{sign}\left(\sum_{j=1}^{12} r_{t-j}\right)$
  - $+1$ (long) if the past 12 monthly log returns sum to a gain, -1 (short) otherwise
  - strategy return $s_t R_t$, R_t the index return of month t; bars: HAC t (6 lags)
- Significant: BET ($t = 2.68$, 1.40% a month), S&P 500 ($t = 2.22$); Bitcoin $t = 1.94$
- Sharpe ratio below buy-and-hold: BET 0.65 vs 0.75, S&P 500 0.48 vs 0.52: much of TSMOM is the equity premium earned while long
  - annualised ($\sqrt{12}\times$ monthly mean/SD), on price returns, with no risk-free rate subtracted: not directly comparable with the monthly Sharpe ratio of the forecasting example
- Data limits: price indices without financing rates or dividends, 15 markets tested; full table: Appendix

MFM_ch2_tsmom

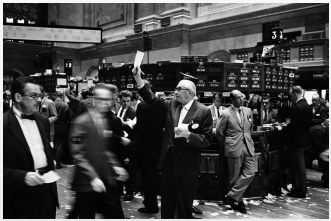
Behavioural concepts: definitions and examples

- Loss aversion gives losses more weight than equal gains relative to a reference point: losing 100 can matter more than gaining 100.
- Prospect theory evaluates gains and losses relative to a reference point, with diminishing sensitivity and probability weighting
  - it is broader than loss aversion.
- Disposition effect is the tendency to sell winners and retain losers: selling a share bought at 100 when it reaches 110, but refusing to sell another at 90.
- Anchoring gives an initial number excessive weight: treating the purchase price 100 as a target even after news changes value.

Beliefs, attention and herding

- Overconfidence exaggerates the precision of one's beliefs: a trader treats a noisy forecast as nearly certain and trades too much.
- Limited attention leaves relevant information unprocessed: an investor reacts to a headline but misses the debt figures in the report.
- Herding means following others' actions: buying because other traders buy, without an independent assessment of value.
- Underreaction adjusts too little to news at first
  - overreaction adjusts too much and is followed by reversal.
  - Neither is established by one isolated price move.

Psychology in Prices, from the Floor to the Feed



New York Stock Exchange trading floor, 1963

Photo: Thomas J. O'Halloran (1963); public domain; Wikimedia Commons

- **Excess volatility** Shiller (1981): prices move far more than the dividends that should justify them; **overreaction**: the source of the 3–5-year reversal De Bondt & Thaler (1985); portraits: ▸ Appendix
- **Prospect theory** Kahneman & Tversky (1979): losses loom larger than gains relative to a reference point
  - Disposition effect: winners sold too early, losers held too long
- **Overconfidence**: too much trading; **anchoring** and **limited attention**: slow reaction to news, a candidate cause of momentum
- **Herding** Pele, Mazurencu-Marinescu & Nijkamp (2013): once voices on a floor, now order books, algorithms and social media (Chapter 10)
- Biases alone do not break efficiency: rational arbitrageurs could trade against them. Can they?

Limits to Arbitrage



Andrei Shleifer, Harvard

Photo: Manuel Castells (2018); CC BY-SA 2.0; Wikimedia Commons

- Shleifer & Vishny (1997): real arbitrage is risky and needs capital
- **Noise-trader risk** De Long, Shleifer, Summers & Waldmann (1990): mispricing can get worse before it corrects
  - Noise traders: investors who trade on signals without information content (sentiment, rumours)
- Funding constraints: managers lose clients after losses and must close positions at the worst time
- Costs: short-selling limits, margins, trading costs, no perfect substitutes
- Anomalies can persist where arbitrage is hard: in small, illiquid markets such as BET we indeed find the strongest persistence
- The AMH reconciles both views: rational and biased behaviour coexist and evolve

Machine learning and research design

- How can a flexible model or research design be evaluated honestly?
- Prerequisite: forecasts on new data
- Topics
- Contents

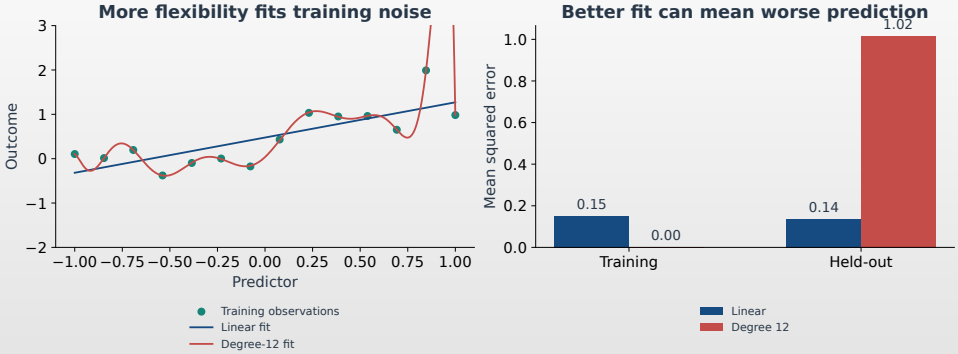
Tools for evaluating an efficiency claim

- Review robust variance-ratio tests.
- Read the definitions of spot ETF and price discovery.
- Review out-of-sample evaluation.

Learning, validation and leakage

- A decision tree splits observations using predictor thresholds
  - a neural network combines weighted nonlinear transformations learned from data.
- Example: a tree can forecast differently when past volatility exceeds 2%
  - this flexibility can also fit accidental sample patterns (overfitting).
- Cross-validation evaluates held-out subsets
  - walk-forward validation respects time, for example training through 2020 and testing 2021.
- Leakage uses information unavailable at prediction time: a revised 2022 accounting figure cannot be used for a forecast made in 2020.

Overfitting: fit versus prediction



Illustrative simulation: the flexible curve fits training noise better but has larger held-out error. Both models use the same training observations.

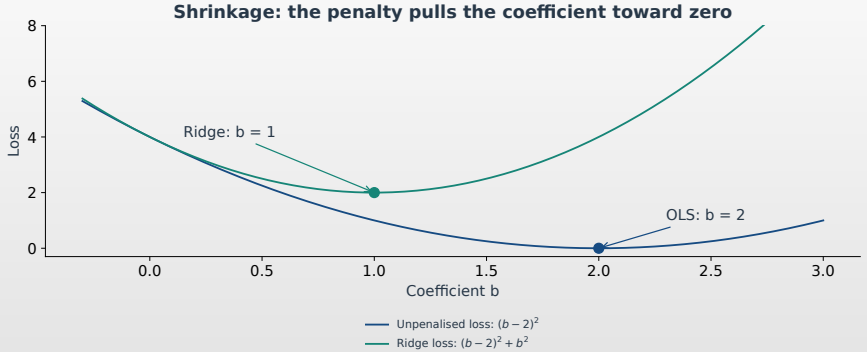
Can Machine Learning Beat Efficient Markets?

- ML (machine learning) learns a prediction rule from data and captures nonlinear patterns that autocorrelation tests miss (Chapter 13)
- Gu, Kelly & Xiu (2020): trees and neural networks give large economic gains over linear regressions in the US cross-section, yet their monthly out-of-sample R^2 is below 1%
- Large language models read news and filings: Tak & Pele (2025) forecast stock trends with LLaMA
- What ML adds to the cautions of this chapter: **chronological validation**
  - train only on data available before each forecast (walk-forward); random cross-validation leaks the future (look-ahead bias)
- Key question: when many predictors are available, what does an in-sample rejection of efficiency mean? The case study answers it

Bayesian learning and shrinkage

- A prior describes parameter uncertainty before new data
  - the likelihood measures how compatible data are with each parameter
  - Bayes' rule combines them into a posterior.
- Illustration: an uncertain prior mean 0 and a data estimate 2 with equal precision give posterior mean 1 in a Normal model.
- Ridge adds a penalty proportional to squared coefficients to the OLS loss
  - shrinkage pulls coefficients toward zero to reduce estimation variance.
- In a one-coefficient example, minimising $(b-2)^2 + b^2$ gives $b = 1$, whereas unpenalised OLS gives 2
  - shrinkage trades bias for lower variance.

Ridge: how the penalty changes the optimum

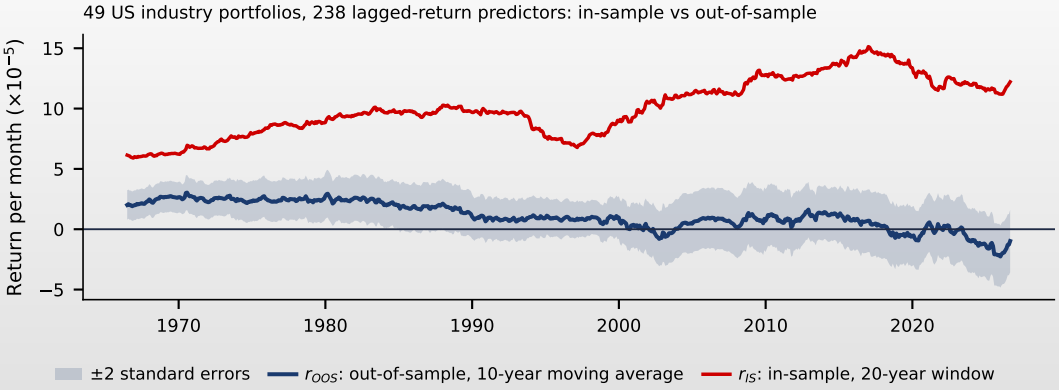


Without the penalty the minimum is at b = 2. Adding the squared-coefficient penalty shifts the minimum to b = 1.

Case Study: Rational Learning with Many Predictors

- Martin & Nagel (2022): investors price N assets whose cash flows depend on J characteristics with **unknown** coefficients and learn them optimally (Bayes) from past data; research question: Appendix
  - the posterior mean is a ridge regression (OLS with a penalty on the squared coefficients): coefficients shrunk towards 0; investors are risk-neutral, so no predictability can be a risk premium
- **In sample**: with $J/N \to \psi > 0$ the test of no predictability rejects with probability tending to 1 (Martin & Nagel (2022), Prop. 4); formal results: Appendix
  - ψ: the limit of the number of predictors per asset, J/N
  - a rejection is not evidence of irrationality: the rational-expectations null is the wrong benchmark when parameters must be learned
- **Out of sample**: a portfolio built with coefficients from other periods has $\mathbb{E}[r_{OOS}] = 0$ (Martin & Nagel (2022), Prop. 6)
  - r_{OOS}: the return of that portfolio in a period not used to estimate its coefficients
  - out-of-sample tests keep their economic meaning: they are the right test of efficiency
- US stocks: in-sample R^2 about 1%, stable; out of sample much lower and within two standard errors of 0 after 2014 (Martin & Nagel (2022), Fig. 8). Model, simulations and design: Appendix

Replication on Course Data: In Sample vs Out of Sample (1/2)



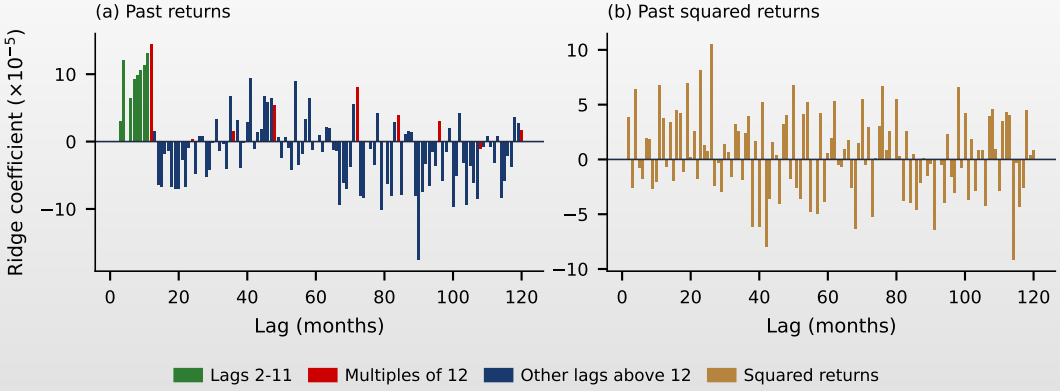
MFM_ch2_big_data

Replication on Course Data: In Sample vs Out of Sample (2/2)

- 49 industry portfolios (Kenneth French Data Library), July 1926 – August 2026, 238 lagged-return predictors, rolling 20-year OLS, specification of Martin & Nagel (2022), Sec. 7
- r_{IS} (in-sample return of the predicted-return portfolio): 5.90 to 15.14 ($\times 10^{-5}$ a month), positive by construction; mean in-sample $R^2 = 5.4\%$
- r_{OOS} (out-of-sample return of the same portfolio), 842 months: mean 1.05×10^{-5}, about 11% of r_{IS}; to June 1986: 4.16 SE above 0, afterwards 0.62
- Bands (± 2 SE) are descriptive, not a test of $\mathbb{E}[r_{OOS}] = 0$; 49 portfolios instead of thousands of stocks make J/N large; details: ▸ Appendix

MFM_ch2_big_data

Replication on Course Data: Ridge Coefficients by Lag (1/2)



MFM_ch2_big_data

Replication on Course Data: Ridge Coefficients by Lag (2/2)

- Full-sample ridge, January 1971 – August 2026, penalty by leave-one-year-out cross-validation (56 years); compare Martin & Nagel (2022), Fig. 7
- Lags 2–11: 8 of 10 positive, **momentum** Jegadeesh & Titman (1993); multiples of 12: 9 of 10 positive, **seasonality** Heston & Sadka (2008)
- Other lags above 12: 67% negative, long-run **reversal** De Bondt & Thaler (1985); squared returns: no dominant sign
- Cross-validated $R^2 = 0.013\%$ (OLS: -2.26%): shrinkage recovers the known anomalies, yet they predict almost nothing in the years left out; limits and extensions: Appendix

MFM_ch2_big_data

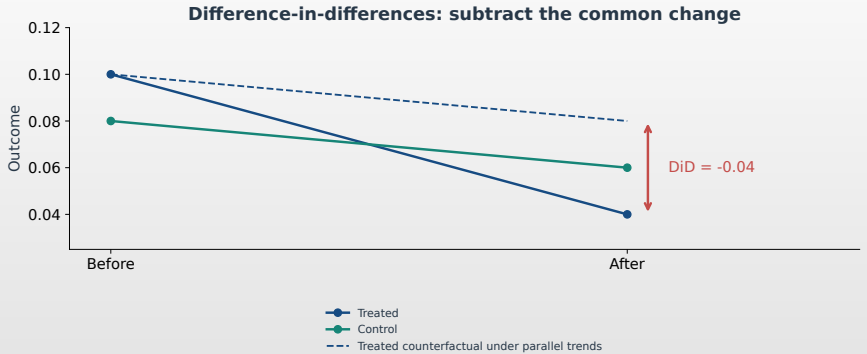
Hypothesis: Did Spot ETFs Make Bitcoin More Efficient?

- A typical assistant proposal: "US spot Bitcoin ETFs (trading since 11 January 2024) brought arbitrageurs, so Bitcoin became more efficient"
- Discovery loop in one pass; reusable prompts: Appendix
  - **literature**: Kia, Liu, Li, Song & Xu (2025) find that after the spot ETFs the spot market became the dominant venue for price discovery again; Tran & Leirvik (2020): crypto efficiency varies and has increased; resolve every DOI yourself
  - **hypothesis**: $H_0 : \rho_{1,\text{post}} = \rho_{1,\text{pre}}$, two windows of equal length around the launch date (first trading day)
  - **data and code**: daily Bitcoin closing price (USD); first replicate the chapter's full-sample $\hat{\rho}_1 = -0.022$
- Checks fixed before seeing results; rationale for each: Appendix
  - robust standard errors (not $1/\sqrt{T}$); equal windows; the launch date only, plus placebo dates
  - a control asset for confounders (the April 2024 halving, rates, inflows)
  - a correction for every statistic tried; the minimum detectable effect

Difference-in-differences: example and assumptions

- DiD subtracts the control group's before–after change from the treated group's change.
- If the treated outcome falls from 0.10 to 0.04 and the control from 0.08 to 0.06, DiD is $(0.04 - 0.10) - (0.06 - 0.08) = -0.04$.
- Parallel trends means the groups would have had the same change without treatment
  - similar pre-trends support but do not prove that counterfactual.
- No anticipation excludes reacting before treatment
  - no spillover excludes treatment affecting the control, for example a Bitcoin ETF also changing Ethereum trading.

Difference-in-differences: reading the contrast



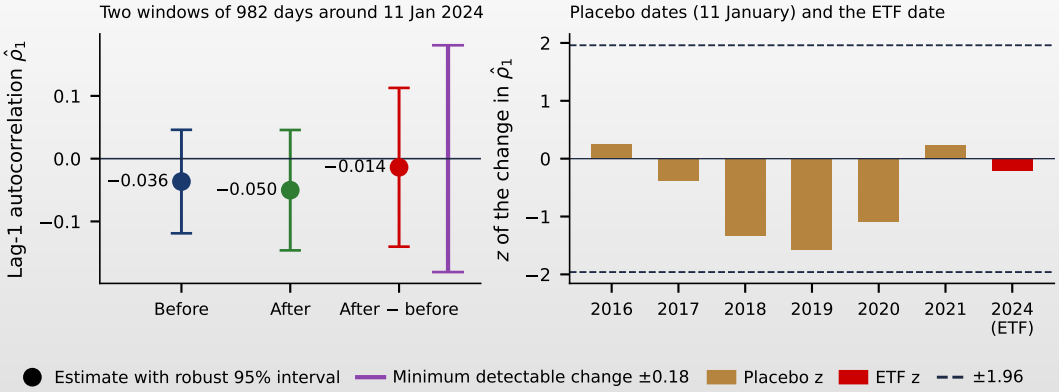
The treated outcome falls by 0.06 and the control by 0.02. The extra fall is 0.04; a causal reading requires the stated assumptions.

Estimation and Power

- Test: two windows of 982 days around 11 January 2024; $z = \Delta\hat{\rho}_1 / \sqrt{s_{\text{pre}}^2 + s_{\text{post}}^2}$, robust standard errors s
  - $\Delta\hat{\rho}_1 = \hat{\rho}_{1,\text{post}} - \hat{\rho}_{1,\text{pre}}$; s_{pre}, s_{post}: the robust SEs of the two estimates, independent windows
  - before (May 2021 – Jan 2024): $\hat{\rho}_1 = -0.036$ ($s = 0.042$), VR(5) $= 1.000$ ($z^* = 0.00$)
  - after (Jan 2024 – Sep 2026): $\hat{\rho}_1 = -0.050$ ($s = 0.049$), VR(5) $= 0.939$ ($z^* = -0.64$)
  - $\Delta\hat{\rho}_1 = -0.014$, $z = -0.21$: an unchanged ρ_1 is not rejected
- **Minimum detectable effect (MDE)**: the smallest true change detected with 80% power at the 5% level
  - MDE $= (z_{0.975} + z_{0.80}) \times \text{SE}(\Delta\hat{\rho}_1) \approx 2.8 \times \text{SE} = 0.18$, z_p: the p-quantile of the standard Normal distribution (1.96 and 0.84)
  - if Bitcoin was already close to efficient before 2024, this design could not detect a smaller improvement
- Placebo at 11 January 2016–2021: 0 of 6 significant, largest $|z| = 1.57$
- Causality would also need a control asset (Ethereum before its own ETFs in July 2024), parallel pre-trends, no anticipation and no spillovers

ai_discovery_case.py

Mini-Case: Estimate, Placebo and MDE (1/2)



ai_discovery_case.py

Mini-Case: Estimate, Placebo and MDE (2/2)

- Left: $\hat{\rho}_1$ before and after, with robust 95% intervals; the difference with $\pm$MDE
- Right: the ETF z lies inside the range produced by six placebo dates
- Placebos load Bitcoin from 2 January 2011; each uses 982 returns before and after its date, including 2016 and 2017.
- "Not significant" is not "no effect": the MDE is several times the estimate
- A better outcome: intraday price discovery across venues, not daily ρ_1

ai_discovery_case.py

Your Turn: A Project Seed

AI: Artificial Intelligence.

- **Question**: did the spot ETFs change the time-varying efficiency of Bitcoin, relative to a control asset?
  - rolling VR and DFA (this chapter) for Bitcoin and Ethereum; Ethereum is untreated until July 2024
  - if parallel trends, no anticipation and no spillovers are not credible, report descriptive, not causal, differences
- **Replicate first**: the chapter's yearly Bitcoin autocorrelations and the mini-case numbers; record which Bitcoin price series and first date each window uses, placebo windows included
- **Pre-register**: H_0, H_1, dates, windows and corrections, committed before the results
- **Report**: robust inference, placebo tests and the MDE; the code as a Quantlet
- **Log**: prompts in `AI_USE.md`; every caught error (an i.i.d. standard error, an invented reference) in `AI_ERRORS.md`

Synthesis, Check Yourself, and Next: Chapter 3

Five steps for every efficiency claim

- **Information set**: which $\mathcal{I}_t$ (past prices, public news)?
- **Null**: RW1, RW3 or the martingale difference, stated together with the risk model
- **Statistic**: its variance valid under volatility clustering
- **Uncertainty**: horizons, markets and windows tried; out-of-sample evidence
- **Economics**: rejection $\neq$ exploitable profit $\neq$ irrational market

Check yourself

- Why can RW3 be rejected for the S&P 500 while the market remains efficient in the economic sense?
- Why does BET's $H(20) > 0.5$ not prove long memory?
- Which of the two results, BET's persistence or Bitcoin's non-rejection, would survive a change of test, window and inference?
- Next, Chapter 3: risk-adjusted returns with factor models, and the factor zoo
- Advanced courses and textbooks: ▸ Appendix

Appendix: Pioneers, Bachelier and Samuelson



Louis Bachelier (1870–1946)

Photo: unknown author; public domain; Wikimedia Commons



Paul A. Samuelson

Photo: Bernard Gotfryd (1970–1975); public domain; Wikimedia Commons

- Bachelier (1900): prices of French government bonds move like a random walk
  - "The mathematical expectation of the speculator is zero"
- Samuelson (1965): if prices are **properly anticipated**, their changes are unpredictable
- Both results are the ancestors of the Efficient Market Hypothesis

Back

Appendix: Eugene Fama



Eugene F. Fama, Nobel Prize 2013

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia Commons

- Fama (1970) gave the definition used in this chapter and the three forms of efficiency
- Nobel Prize 2013, shared with Robert Shiller (a sceptic of efficiency) and Lars Peter Hansen
- Update and survey: Fama (1991); accessible critique: Malkiel (2003)

Back

Appendix: Summary Table, Weak-Form Tests 2000–2026

Market	T	$\hat{\rho}_1$	robust SE	$\widehat{\mathrm{VR}}(5)$	$z^*(5)$	CD p	Runs z
S&P 500 (US)	6 714	−0.099	0.027	0.830	−2.77	0.001	4.34
Euro Stoxx 50	6 836	−0.029	0.019	0.910	−2.08	0.097	3.41
Nikkei 225 (Japan)	6 545	−0.038	0.024	0.923	−1.48	0.317	1.97
Hang Seng (HK)	6 583	−0.004	0.024	0.967	−0.64	0.872	−0.17
BET (Romania)	6 670	0.109	0.028	1.173	2.98	0.001	−6.05
WIG20 (Poland)	6 685	0.018	0.018	1.009	0.24	0.777	2.25
BUX (Hungary)	6 666	0.037	0.024	1.013	0.26	0.427	−0.42
PX (Czechia)	6 682	0.052	0.032	1.041	0.59	0.350	−2.53
BIST 100 (Turkey)	6 692	0.005	0.024	1.033	0.64	0.949	−0.63
Bovespa (Brazil)	6 622	−0.028	0.025	0.928	−1.29	0.521	0.88
IPC (Mexico)	6 689	0.070	0.019	1.039	0.93	0.001	−3.27
Nifty 50 (India)	6 606	0.048	0.024	1.048	0.91	0.178	−5.46
Shanghai (China)	6 474	0.021	0.019	1.046	1.11	0.409	1.02
Bitcoin	4 384	−0.022	0.026	0.991	−0.19	0.835	2.68
Ethereum	3 913	−0.012	0.027	1.067	1.17	0.027	3.91
EUR/RON (reference rate)	5 352	0.170	0.036	1.157	2.19	0.000	−1.66

- T: number of daily returns; CD: Chow–Denning over $q = 2, 5, 10, 20$; EUR/RON from July 2005, crypto on 365 days per year

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Appendix: Runs Test on All Markets, Other VR Tests

- Runs z (table on the previous slide): Nifty 50 -5.46, IPC -3.27 (persistence in emerging markets); Ethereum 3.91, significant although $\hat{\rho}_1$ is not
- Choi (1999): an automatic VR test that chooses the horizon from the data (spectral estimate at frequency zero)
- Wild-bootstrap versions for small samples: Kim (2006); applications to Asian markets: Kim & Shamsuddin (2008)

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Appendix: choosing lags with AIC and BIC

- **AIC**: Akaike Information Criterion.
- **BIC**: Bayesian Information Criterion.
- A penalty discourages adding lags merely because fit improves by chance
  - the automatic test balances fit and complexity.
- In the criterion below, AIC uses penalty $2p$ and BIC uses $p \ln T$ for p included lags and T observations.
- At $T = 100$, adding one lag costs 2 under AIC and about 4.61 under BIC
  - a small fit gain will not justify the extra lag.

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Appendix: Automatic Portmanteau Test of the MDH

Prerequisite: AIC and BIC

- Escanciano & Lobato (2009): $AQ = T \sum_{j \le \tilde{p}} \hat{\gamma}_j^2 / \hat{\kappa}_j \to \chi^2_1$, with $\hat{\gamma}_j = \frac{1}{T-j} \sum_t e_t e_{t-j}$ and $\hat{\kappa}_j = \frac{1}{T-j} \sum_t e_t^2 e_{t-j}^2$ (robust to conditional heteroskedasticity)
  - $\tilde{p} \le 10$ maximises $Q_p = T \sum_{j \le p} \hat{\gamma}_j^2 / \hat{\kappa}_j$ minus a penalty that switches from BIC ($p \ln T$) to AIC ($2p$) when some $\sqrt{T}|\hat{\rho}_j|$ is large
- $e_t = r_t - \bar{r}$; $\hat{\gamma}_j$: autocovariance at lag j; $\hat{\kappa}_j$: its robust variance; a large AQ rejects the MDH
- AQ: S&P 500 13.1 ($p = 0.0003$), BET 14.9 ($p = 0.0001$), Bitcoin 0.8 ($p = 0.385$); the MDH is rejected at 5% for 5 of the 16 series

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Appendix: Predictive Regression, Data and Full Output

FRED: Federal Reserve Economic Data.

- Monthly log excess return (dividends included, minus the 3-month T-bill rate) regressed on the lagged dp_t
  - Data: Shiller (2026) (S&P Composite, to June 2026), extended with monthly averages of our daily S&P 500 closes (scaled over the last 12 common months) and the dividends of the SPY fund; 3-month US Treasury bill yield (FRED, series TB3MS; annual %), monthly log rate $\ln(1 + \text{TB3MS}/1200)$ lagged one month
- OLS: $\hat{\beta} = 0.267$, Newey–West SE 0.293 (12 lags), $t = 0.91$, OLS $t = 1.16$, $R^2 = 0.12\%$
- Corrected slope / unadjusted HAC SE $= -0.22$: only illustrative, it ignores the uncertainty of the correction
- Economic value Campbell & Thompson (2008) (number: Out-of-Sample Evidence slide): proportional rise in the expected return of the market timer
  - $\dfrac{R^2}{1-R^2} \cdot \dfrac{1+S^2}{S^2}$, S the monthly Sharpe ratio
- Diebold–Mariano Diebold & Mariano (1995) $t = -0.57$ for the same forecasts, against Clark–West $t = 0.42$

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Appendix: Harold Hurst and the Nile



Harold Edwin Hurst (1880–1978)

Photo: Elliott & Fry (1953); public domain; Wikimedia Commons



Nilometer on Elephantine Island, Aswan

Photo: Olaf Tausch (2011); CC BY 3.0; Wikimedia Commons

- Hurst (1951) designed Nile reservoirs; his finding and the **rescaled range**: R/S slide

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Appendix: the gamma function

- The gamma function extends factorials: $\Gamma(x) = \int_0^\infty t^{x-1} e^{-t}\, dt$ for $x > 0$.
- For integer $n \geq 1$, $\Gamma(n) = (n-1)!$
  - for example $\Gamma(4) = 6$, while $\Gamma(1/2) = \sqrt{\pi}$.
- These values supply the normalising constants in the exact Anis–Lloyd mean
  - the expression is a benchmark under i.i.d. Normal returns, not a fitted trend.

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Appendix: The Anis–Lloyd Expectation

Prerequisite: the gamma function

- Anis & Lloyd (1976): exact $\mathbb{E}[R_n/s_n]$ under i.i.d. returns from the Normal distribution
  - $\mathbb{E}[R_n/s_n] = \dfrac{\Gamma((n-1)/2)}{\sqrt{\pi}\,\Gamma(n/2)} \sum_{i=1}^{n-1} \sqrt{\dfrac{n-i}{i}}$ (Γ: the gamma function)
- This $\mathbb{E}[R_n/s_n]$ is the reference subtracted in H_{AL} (R/S Analysis slide)
- Uncorrected estimates overstate memory by about 0.04–0.05 with daily data

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Appendix: Long Memory with Inference, the Local Whittle Estimator

- $d = H - 0.5$; spectral density $f(\lambda) \sim G\lambda^{-2d}$ as $\lambda \to 0$ ($G > 0$ a constant; $d > 0$: the density explodes at low frequencies, long memory); fit only the first m Fourier frequencies $\lambda_j = 2\pi j/T$
  - $\hat{d} = \arg\min_d \left[\ln\left(\frac{1}{m} \sum_{j=1}^{m} \lambda_j^{2d} I(\lambda_j) \right) - \frac{2d}{m} \sum_{j=1}^{m} \ln \lambda_j \right]$, I: the periodogram
  - Robinson (1995): for $|d| < 1/2$, $m \to \infty$, $m/T \to 0$, a smooth f near 0 and a linear process with conditionally homoskedastic martingale-difference innovations, $\sqrt{m}(\hat{d} - d) \to N(0, 1/4)$ without Gaussianity: a confidence interval instead of a band
- Estimates with $m = T^{0.65}$ and the 95% CI $\hat{d} \pm 1.96/(2\sqrt{m})$
  - S&P 500 -0.013 $[-0.069, 0.043]$; Bitcoin 0.063 $[-0.002, 0.127]$
  - BET 0.118 $[0.062, 0.174]$; Ethereum 0.112 $[0.045, 0.179]$
- Bandwidth trade-off: a large m reaches frequencies where short-run autocorrelation (BET $\hat{\rho}_1 = 0.109$) biases $\hat{d}$ upward
  - with $m = T^{0.5}$ BET still gives 0.135 (SE 0.056): a sensitivity check supporting, not proving, that its $d > 0$ is not only short memory
  - GARCH returns also strain the conditions above

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Appendix: EMH versus AMH

	EMH	AMH
Investors	Rational, optimising	Bounded rationality, heuristics
Efficiency	An equilibrium property, tested as a restriction	A matter of degree that changes with conditions
Risk–return	Equilibrium risk premia, possibly time-varying	Changes with the population of investors
Anomalies	Omitted risk factors or data mining	Transient profit opportunities
Empirical tool	Tests of an equilibrium restriction	Tests of time and state variation (rolling windows)

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Appendix: Is Bitcoin Efficient? The Literature

- Urquhart (2016): Bitcoin inefficient over 2010–2016; some tests indicate efficiency in the later subperiod
  - His battery (Ljung–Box, runs, the rank test of Bartels (1982), automatic VR, BDS, R/S) includes BDS, which rejects under volatility clustering alone: not a test of RW3
- Nadarajah & Chu (2017): with a power transformation of returns, Bitcoin is weak-form efficient
- Kristoufek (2018): Bitcoin markets mostly inefficient in 2010–2017, with efficiency changing over time
- Tran & Leirvik (2020): time-varying efficiency across many crypto-assets, increasing over time
- Pele et al. (2023): crypto-assets have become a distinct asset class; their returns share the stylised facts of other markets
- Our own estimates, calendar year by calendar year (slide Crypto Efficiency Year by Year)
  - Bitcoin, lag-1 autocorrelation $\hat{\rho}_1$: -0.002 in 2011, -0.071 in 2012, 0.069 in 2013, all close to zero
  - Robust variance-ratio statistic $z^*(5)$: at most 1.71 in absolute value over all crypto-asset years, below the critical value 1.96
  - So in the early years we do not reject RW3, unlike Urquhart (2016)

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Appendix: Romania, Frontier then Emerging

BVB: Bursa de Valori București — the Bucharest Stock Exchange.

- FTSE Russell (2019): Romania reclassified from Frontier to Secondary Emerging, announced in September 2019, effective 21 September 2020
- Expected channels: more foreign investors, more analysts, more liquidity $\Rightarrow$ faster price discovery
- Earlier BVB evidence: crashes and log-periodic bubbles Pele & Mazurencu-Marinescu (2012); herding in an illiquid market Pele, Mazurencu-Marinescu & Nijkamp (2013)
- Did efficiency improve? Seminar C2 compares five years before and after, with WIG20 and BUX as a comparison group and a block bootstrap
  - Block bootstrap: resamples blocks of consecutive days to keep the dependence Künsch (1989); a before/after comparison is descriptive unless confounders are controlled

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Appendix: How Fast Is "Fast"?

CME: Chicago Mercantile Exchange.

- At high frequency, futures prices react to scheduled macro news within microseconds; Tak, Schlamp, Osterrieder & Pele (2026) measure this race at Eurex and CME
- Crypto events: Pele, Lin, Wang & Gaman (2025) study market responses to Ethereum development milestones
- Speed of adjustment is itself a measure of efficiency, and it depends on technology (Chapter 10, market microstructure)

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Appendix: Reality Check, SPA and StepM explained

SPA: Superior Predictive Ability (test).

- Reality Check asks whether the best of all tried rules beats a benchmark, using a bootstrap distribution of their maximum performance. White (2000)
- SPA standardises performance by uncertainty (studentisation) and reduces distortion from poor rules
  - a gain 2 with SE 1 gives a studentised statistic 2. Hansen (2005)
- StepM tests the strongest statistic first, then the remaining family using jointly resampled critical values
  - it controls FWER under its assumptions. Romano & Wolf (2005)
- Example: resample the same months for all 15 calendar rules, retaining common market shocks
  - separate resampling would destroy that dependence.

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Appendix: Dependent Tests, Reality Check, SPA and StepM

Plain-language version

- Bonferroni and Holm ignore the correlation between tests (same markets, same days)
  - White (2000): Reality Check, bootstrap of $\max_j$ over all rules tried; Hansen (2005): SPA, studentised and less sensitive to poor rules
  - Romano & Wolf (2005): StepM (Romano–Wolf step-down procedure), step-down control of the FWER from the **joint** bootstrap distribution of all statistics
- Our 15 calendar tests: StepM with a bootstrap of blocks of 3 calendar months, common to all markets, 999 replications
  - Critical p at the first step 0.0016 (Bonferroni: 0.0033); 2 rejections, BET and BUX TOM, as with Holm
  - Why stricter than Bonferroni? At nominal 5%, the bootstrap rejection rates of the HAC tests are 6.0% (day of week), 6.4% (TOM), 6.4% (January)
- From over two million simulated strategies: thresholds $|t| > 3.8$ (time series) and 3.4 (cross-section); ignoring them makes about 45% of rejections false Chordia, Goyal & Saretto (2020); lowering false discoveries raises missed ones Harvey & Liu (2020)

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Appendix: Time-Series Momentum, All Markets

Market	Mean (%/month)	t_{HAC}	Sharpe TSMOM	Sharpe buy-and-hold	$\hat{\rho}_1$ monthly
S&P 500 (US)	0.61	2.22	0.48	0.52	0.03
Euro Stoxx 50	0.16	0.51	0.11	0.16	0.05
Nikkei 225 (Japan)	0.61	1.85	0.39	0.42	0.09
Hang Seng (HK)	0.18	0.53	0.10	0.19	0.04
BET (Romania)	1.40	2.68	0.65	0.75	0.19
WIG20 (Poland)	0.67	1.86	0.36	0.26	0.02
BUX (Hungary)	0.72	1.79	0.40	0.64	0.10
PX (Czechia)	0.38	0.99	0.24	0.45	0.10
BIST 100 (Turkey)	1.03	1.85	0.37	0.75	−0.08
Bovespa (Brazil)	0.28	0.65	0.14	0.51	0.11
IPC (Mexico)	0.44	1.41	0.32	0.63	0.03
Nifty 50 (India)	0.50	1.29	0.28	0.64	0.02
Shanghai (China)	0.59	1.32	0.30	0.23	0.08
Bitcoin	4.47	1.94	0.72	1.08	0.13
Ethereum	6.50	1.28	0.57	0.90	0.16

- 307 months for equities, fewer for crypto; Sharpe ratio $\sqrt{12}\,\bar{x}/s_x$ of price returns ($\bar{x}$, s_x: mean and SD of the monthly strategy returns), no risk-free rate subtracted; Moskowitz, Ooi & Pedersen (2012) use futures excess returns scaled to 40% ex-ante volatility

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Appendix: Robert Shiller and Richard Thaler



Robert J. Shiller, Nobel Prize 2013

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia Commons



Richard H. Thaler, Nobel Prize 2017

Photo: Bengt Nyman (2017); CC BY 2.0; Wikimedia Commons

- Excess volatility Shiller (1981) and overreaction De Bondt & Thaler (1985): Psychology in Prices slide

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Appendix: Case Study, Research Question

- Martin & Nagel (2022): investors price N assets whose cash flows depend on J characteristics with unknown coefficients; what does a rejected no-predictability test mean when J is comparable to N?
  - the coefficients must be learned from past data: a high-dimensional prediction problem, as in machine learning
- Semi-strong efficiency: prices "fully reflect" public information Fama (1970); two readings
  - investors know the parameters of the cash-flow model (rational expectations: the null of most efficiency tests)
  - investors update their beliefs optimally (Bayes) with the data available in real time
- Why it matters: hundreds of in-sample return predictors, the "factor zoo" Harvey, Liu & Zhu (2016); text, satellite images and social media keep increasing J
  - does every significant predictor need a risk-based or a behavioural explanation?
- Link to this chapter: in-sample vs out-of-sample predictability Campbell & Thompson (2008), Goyal, Welch & Zafirov (2024), and the decay of anomalies after publication McLean & Pontiff (2016)

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Appendix: Case Study, Model and Method (1/2)

- **Model** (Martin & Nagel (2022), Sec. 2)
  - cash-flow growth $\Delta y_t = Xg + e_t$, with $e_t \sim N(0, I)$ and a zero interest rate
  - Δy_t: the N cash-flow growth rates; X: the $N \times J$ matrix of characteristics; g: the J unknown coefficients; I: identity matrix
- **Learning** (Prop. 1–2)
  - prior $g \sim N(0, (\theta/J)\, I)$, the true distribution from which g is drawn; θ: signal strength
  - realised return = underreaction caused by shrinkage + estimation noise + unpredictable shock
- **Asymptotics and simulations** (Sec. 4)
  - $N, J \to \infty$ with $J/N \to \psi > 0$
  - simulations with $N = 1000$, entries of X drawn from the standard Normal distribution and $\theta = 1$

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Appendix: Case Study, Model and Method (2/2)

- **CRSP**: Center for Research in Security Prices.
- **NYSE**: New York Stock Exchange.
- **Tests**
  - in sample: a cross-sectional OLS of r_{t+1} on X (requires $J < N$) with a joint χ^2 test
  - out of sample: a portfolio with weights $\frac{1}{N} X h_{s+1}$, from coefficients estimated in other periods
  - $r_{\text{OOS},t+1} = \frac{1}{N} h'_{s+1} X' r_{t+1}$, $s \neq t$ (eq. 19); r_{IS}: the same with $s = t$
  - h_{s+1}: the estimated coefficient vector; r_{t+1}: the N returns
- **Empirical design** (Sec. 7)
  - US stocks in CRSP, without those below the 20th NYSE size percentile or priced below \$1
  - predictors: simple and squared monthly returns at lags $t-2$ to $t-120$ (238 predictors), demeaned and standardised across stocks every month
  - full-sample ridge, 1971 – June 2019, penalty chosen by leave-one-year-out cross-validation
  - rolling 20-year OLS gives r_{OOS} from January 1956

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Appendix: Case Study, Formal Results

- In sample (Prop. 4 of Martin & Nagel (2022), main slide): for $\psi > 0.4$ ($\theta = 1$, one period of learning) the probability of no rejection is of the order $e^{-0.015N}$ (Fig. 2)
- Simulations ($N = 1000$, four periods of learning): a test of nominal size 5% rejects with probability above 90% once $J > N/2$ (Fig. 3b)
  - not a size distortion: the in-sample predictable component really is in the prices
- Single-predictor tests with k-FWER control (probability of k or more false rejections at most 5%): for $k = 20$ and $J/N = 0.05$, the probability of at least k rejections is 25% instead of 5%; about 80% for J/N close to 1 (Fig. 4)
  - multiple-testing corrections do not help: the problem is the null hypothesis, not the test
- Out of sample: $\mathbb{E}[r_{\mathrm{OOS},t+1}] = 0$ holds forward and backward in time (Prop. 6); positive only if investors shrink more than the data warrant (Fig. 6)
- US stocks: the ridge coefficients recover momentum, long-run reversal and momentum seasonality (Fig. 7); $\eta_S \approx 0.006\%$ per month (Fig. 8)

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Appendix: Case Study, Replication, Limitations and Extensions

- **Replicate with the course data** (stock-level CRSP files are not part of them)
  - the 49 value-weighted industry portfolios of the main replication slides (Seminar 2, C3); same predictors, standardisation and windows
  - the size and price filters do not apply to portfolios
- **Limits stated by the authors**
  - lagged returns are equilibrium outcomes, not the exogenous X of the model: the application illustrates the wedge, it does not test the model
  - a t-test on the monthly r_{OOS} is not a valid test of $\mathbb{E}[r_{OOS}] = 0$: the expectation also averages over g
  - the backward result may rely on i.i.d. cash-flow growth; homogeneous investors and constant characteristics
- **Open extensions**
  - time series, as the authors suggest: does the dividend–price regression of this chapter show the same in-sample/out-of-sample wedge?
  - nonlinear models (trees, neural networks Gu, Kelly & Xiu (2020)) add functional flexibility without necessarily changing J: does the wedge grow?
  - AMH: is early out-of-sample predictability the excess shrinkage of investors with less computing power?

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Appendix: Case Study, In Sample vs Out of Sample in Detail

- $r_{\text{IS}} \geq 0$ in every 20-year window by construction: $r_{\text{IS}} = b' G^{-1} b/240$, with $G = \sum_s X_s' X_s / N_s$ and $b = \sum_s X_s' r_s / N_s$ over the 240 months s of the window
  - X_s: the $N_s \times 238$ predictor matrix of month s; r_s: the N_s portfolio returns; $G^{-1}b$: the pooled OLS coefficients
  - mean in-sample R^2 of 5.4%, against about 1% for individual stocks in Martin & Nagel (2022): 238 predictors against only 40–49 portfolios a month, pooled over 240 months, inflate the in-sample fit
- Out of sample (from July 1956), until June 1986: mean $r_{\text{OOS}} = 1.98 \times 10^{-5}$, 4.16 standard errors above 0; afterwards 0.36×10^{-5}, only 0.62 standard errors
  - the 10-year average is last above two standard errors in the window ending in December 1987; in the windows ending in 2014–2019 its largest ratio to the standard error is 1.25, as for stocks
- Interpretation suggested by Martin & Nagel (2022): the gap is consistent with investor learning, but does not establish it or rule out risk premia; early out-of-sample gains are also consistent with excess shrinkage before the computer era (Fig. 6)

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Appendix: The Discovery Loop, Reusable Prompts (1/2)

- **Literature**
  - `List peer-reviewed studies since 2020 on how ETF introductions change the efficiency or price discovery of the underlying. For each: DOI, design (event study, difference-in-differences (DiD), structural break), efficiency measure. Flag any entry you are unsure exists.`
  - Resolve every DOI on Crossref; the title and authors must match
- **Hypothesis**
  - `Give three mechanisms by which spot ETFs could change Bitcoin's serial dependence (arbitrage, investor mix, trading hours). For each, the sign of the change in rho1 and VR(q), and a variable that separates it from the others.`
- **Data and code**
  - `Read the daily Bitcoin closing prices from the course data; log returns; rho1 with a heteroskedasticity-robust standard error and the Lo-MacKinlay VR(5) with z*, in two windows of equal length around 11 January 2024.`

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Appendix: The Discovery Loop, Reusable Prompts (2/2)

- **Robustness**
  - `Write a pre-registered robustness plan: window lengths, weekday-only closes, VR(q) for q = 2 to 20 with Chow-Denning, placebo dates, and Ethereum as a control until its own spot ETFs (July 2024).`
- **Critique**
  - `Act as a hostile referee: if Bitcoin was already efficient before 2024, what could this test ever detect? Compute the minimum detectable change in rho1 and name a better outcome variable.`
- **Report**
  - Estimates, robust standard errors, the placebo distribution, every test run (with a multiple-testing correction)
  - Prompts that shaped a decision go into `AI_USE.md`
- The assistant drafts the tests; the human decides which one answers the question

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Appendix: What the Researcher Must Check

- **i.i.d. standard errors**: $1/\sqrt{T}$ rejects too often under volatility clustering
  - test: robust SE and z^*, as in the weak-form tests of this chapter
- **Unequal windows**: different power before and after; test: equal lengths, fixed in advance
- **Event date chosen from the data**
  - test: the launch date (first trading day) only, plus a placebo distribution at dates without an event
- **Confounding**: before/after is not causal; test: a control asset (Ethereum before July 2024) in a difference-in-differences
  - causal only under the conditions of the Estimation and Power slide; the common post-period (Jan–Jul 2024) is short
- **Many statistics**: ρ_1, VR(2–20), runs; test: Chow–Denning or Holm, not the smallest single p-value
- **Low power**: report the MDE of the Estimation and Power slide

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Appendix: Further Study on Quantinar

Advanced courses on quantinar.com

- Efficiency of cryptocurrency markets - A GMM-based analysis
- Generalized Method of Moments

Code, data and textbooks

- All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_02
- Textbooks: Campbell, Lo & MacKinlay (1997), Ch. 2; Franke, Härdle & Hafner (2019)

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Acronyms used in this material (1/2)

- **AIC**: Akaike Information Criterion
- **AMH**: Adaptive Markets Hypothesis
- **AR(p)**: AutoRegressive model of order p.
- **AR**: Abnormal Return, in event studies.
- **BDS**: Brock–Dechert–Scheinkman (test)
- **BET**: Bucharest Exchange Trading (index)
- **BIC**: Bayesian Information Criterion
- **BIST**: Borsa İstanbul (BIST 100 index) — *the Istanbul Stock Exchange (BIST 100 index)*
- **BNR**: Banca Națională a României — *the National Bank of Romania*
- **BUX**: Budapesti Értéktőzsde index — *the Budapest Stock Exchange index*
- **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- **CAPM**: Capital Asset Pricing Model
- **CAR**: Cumulative Abnormal Return
- **CD**: Chow–Denning (multiple variance-ratio test)
- **CI**: Confidence Interval
- **CME**: Chicago Mercantile Exchange
- **CRSP**: Center for Research in Security Prices
- **DFA**: Detrended Fluctuation Analysis
- **EMH**: Efficient Market Hypothesis
- **ETF**: Exchange-Traded Fund
- **fGn**: fractional Gaussian noise
- **FRED**: Federal Reserve Economic Data
- **FTSE**: Financial Times Stock Exchange (FTSE Russell, index provider)
- **FWER**: Family-Wise Error Rate
- **FX**: Foreign eXchange

Acronyms used in this material (2/2)

- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GMM**: Generalized Method of Moments
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HK**: Hong Kong.
- **IPC**: Índice de Precios y Cotizaciones — *the Mexican Stock Exchange index*
- **MDE**: Minimum Detectable Effect
- **MDH**: Martingale Difference Hypothesis
- **MIT**: Massachusetts Institute of Technology
- **ML**: Machine Learning
- **MSPE**: Mean Squared Prediction Error
- **NYSE**: New York Stock Exchange
- **OLS**: Ordinary Least Squares
- **PX**: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- **RW1**: Random Walk 1 (i.i.d. increments)
- **RW2**: Random Walk 2 (independent increments)
- **RW3**: Random Walk 3 (uncorrelated increments)
- **SD**: Standard Deviation
- **SE**: Standard Error
- **SPA**: Superior Predictive Ability (test)
- **TOM**: Turn Of the Month
- **TSMOM**: Time-Series MOMentum
- **US**: United States.
- **VR**: Variance Ratio
- **WIG20**: Warszawski Indeks Giełdowy 20 — *the Warsaw Stock Exchange index of the 20 largest companies*

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